SAMSUNG ELECTRONICS

Knox E-FOTA On-Premises

Installation and Initial Operation Guide

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PART I: Getting Started

PART 1: Getting Started presents the purpose of this document, the customer infrastructure that is recommended prior to the installation of the Knox E-FOTA On-Premises service, and provides an overview of deliverables to be used during the installation.

1. Introduction

1.1. Purpose of this document

The purpose of this document is to present how to plan for, install, and configure the managed DFM module within the customer's network. This document includes information about how to install and configure 3rd party software, such as Podman, and provides detailed descriptions of the commands used to perform its installations.

This document is intended **for the personnel who are in charge of performing the installation**.

To prepare the installer, this document includes the following tasks:

- Evaluate the customer's network and hardware facilities
- Introduce which modules will be installed to provide this service
- Explain the install flow with DFM Modules
- Explain how to configure the installed DFM Modules with the proper conditions
- Explain how to test if the installed DFM Modules are running as expected

The server infrastructure, hereafter referred to as **DFM Modules** will be installed on the customer's side by Samsung to service the Knox E-FOTA On-Premises environment.

We recommend "The 4-Days Installation" method for this installation, as the customer should understand how they are using this service during this program (see "*Appendix D. A Recommended Schedule for On-Site Installation by CSO/TEO*").

2. Environment Prerequisites

This chapter presents the hardware, software and network facilities required by the DFM. To ensure proper support of E-FOTA On-Premise, the service must be installed upon the following recommended software and hardware infrastructure.

The following recommended items should be prepared by the customer prior to the installation of the Knox E-FOTA On-Premise service by Samsung personnel.

2.1. Recommended Hardware

Table 2-1 below outlines the recommended user environment, including network card for the On-Premise Hardware (H/W) requirements. The customer can choose the correct value depending on the product type—see "***2.3 Recommendation Per each Product Usage***".

Items	Recommended Value	Description
Server CPU Cores	Above 4 or 8 CPU Cores	4 Cores is for PoC Product Above 8 Cores is for Commercial Product
RAM	16 or 32 GB	16GB is for PoC Product 32GB is for Commercial Product
Disk	1TB or 2TB SSD	For DFM Module 1TB (PoC), 2TB (Commercial Product)
	256 GB	For System region (OS and Root filesystem)
Network Card	Above 10 Gbps	

Table 2-1 The recommended hardware for a user work environment to run Knox E-FOTA On-Premises

The recommendations in the above table are the minimum specifications to run the Knox E-FOTA On-Premises service. User performance expectations may require additional infrastructure resources in excess of the minimum specifications.

2.2. Recommended Software

The recommended user work environment, including network, for the On-Premise Software (S/W) requirements are as follows:

Items	Recommended Value	Description
Operating System	Red Hat Enterprise Linux 8.4, 9.2, or 9.6	
Container Tool	Podman	
MySQL Edition	Community Edition	

Table 2-2 The recommended hardware for a user work environment to run Knox E-FOTA On-Premises

Refer to "*Appendix C*" for a summary of software recommendations.

2.2.1. Operating System

By default, the DFM Server requires Red Hat Enterprise Linux 8.4, 9.2, or 9.6 for the OS. It should be installedon 64-bit Intel x86, ARM, or MIPS architectures in order to support Podman.
Selinux mode in the Red hat OS supports permissive and enforcing modes, and the DFM Server can

be installed in each mode.

2.2.2. Podman

Podman is a daemonless container engine for developing, managing, and running OCI Containers on your Linux System. Containers can either be run as root or in rootless mode. Since Red Hat Enterprise Linux 8, Podman is officially supported. For Knox E-FOTA On-Premises, Podman is the recommended container tool for Red Hat users. Podman installation is described in **Section 4.1.**

2.2.3. Database (MySQL)

The MySQL database contains all service-related data, including device models, their IDs, and policy dependencies in Campaigns.

2.2.4. HTTPS

To use the https protocol between Samsung mobile devices and the DFM Modules, the customer should prepare a DNS hostname (FQDN) and public (or private) SSL certificates.

2.3. Recommendation Per each Product usage

Knox E-FOTA On-Premises has 3 types of product use case architecture recommendations, including 2 Commercial and 1 POC architecture.

2.3.1. Product – "PoC"

The **PoC** product is recommended if a customer wants to use the on-premises service to understand its functions and product configuration clearly prior to purchasing a Commercial Product, along with a small number of devices (clients, until 300 devices). The PoC product can run on lower specification hardware than the Commercial product, but the table below contains the minimum specifications to be running Knox E-FOTA On-Premises. To ensure the service runs as expected, the customer should set up the infrastructure with higher specifications than shown below.

【Minimum Hardware Recommendation】

Items	Recommended Value	Description
Server CPU Cores	4 CPU Cores	
RAM	16 GB	
Disk	1TB SSD	For DFM Module
	256 GB	For System region (OS and Root filesystem)
Network Card	Above 10 Gbps	

Table 2-3 The Minimum Hardware Recommendation for PoC

【Software Recommendation】

Items	Recommended Value	Description
Operating System	Red Hat Enterprise Linux 8.4, 9.2, or 9.6	
Container Tool	Podman	
MySQL Edition	Community Edition	

Table 2-4 The Software Recommendation for "PoC"

The customer can purchase Red hat OS based on this service package, depending on their service environment. Note that the customer must provide the service infrastructure to the Samsung representative in charge of the installation.

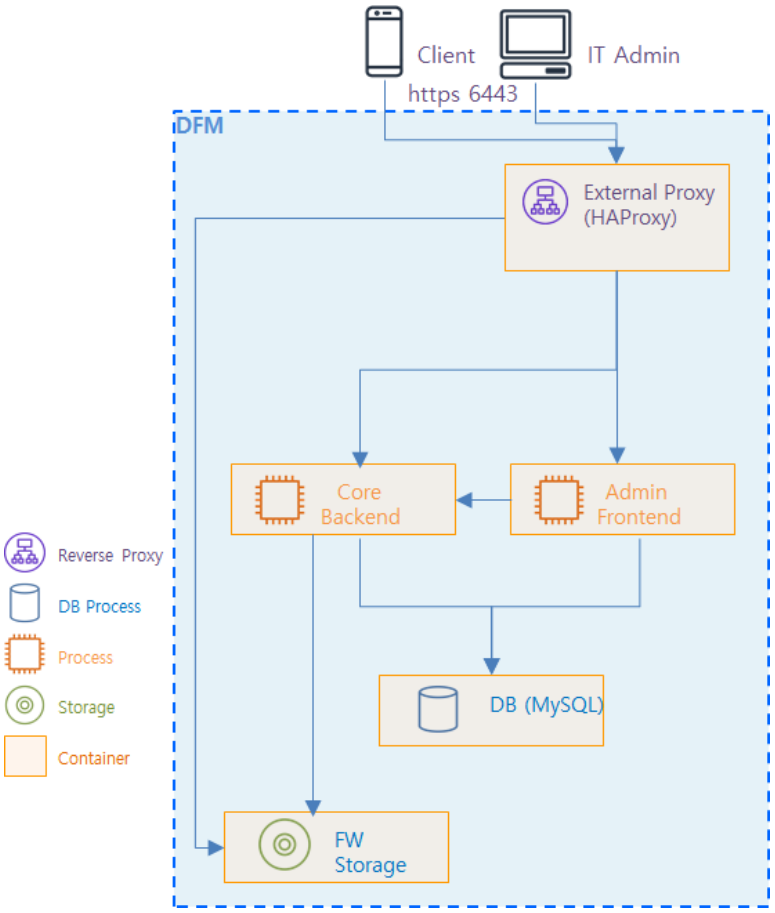


Fig 2-1 Knox E-FOTA On-Premises Product Arch for "PoC"

2.3.2. Product – "Commercial"

The **Commercial** product is recommended for customers who want to use this product with a maximum of 20,000 devices for device firmware updates over-the-air (FOTA), but it also supports more than 20,000 devices.

The recommended specification for the infrastructure is the minimum required to be running the service. To optimize performance expectations, the customer may need to provide infrastructure with higher specifications than the below table to the Samsung representative in charge of the installation.

【Minimum Hardware Recommendation】

Items	Recommended Value	Description
Server CPU Cores	8 CPU Cores	
RAM	32 GB	
Disk	2TB SSD	For DFM Module
	256 GB	For System region (OS and Root filesystem)
Network Card	Above 10 Gbps	

Table 2-5 The Minimum Hardware Recommendation for "Commercial"

【Software Recommendation】

The customer can purchase Red hat OS based on this service package, depending on their service environment. Note that the customer must provide the service infrastructure to the Samsung representative in charge of the installation.

Items	Recommended Value	Description
Operating System	Red Hat Enterprise Linux 8.4, 9.2, or 9.6	
Container Tool	Podman	
MySQL Edition	Community Edition	

Table 2-6 Software Recommendation for "Commercial"

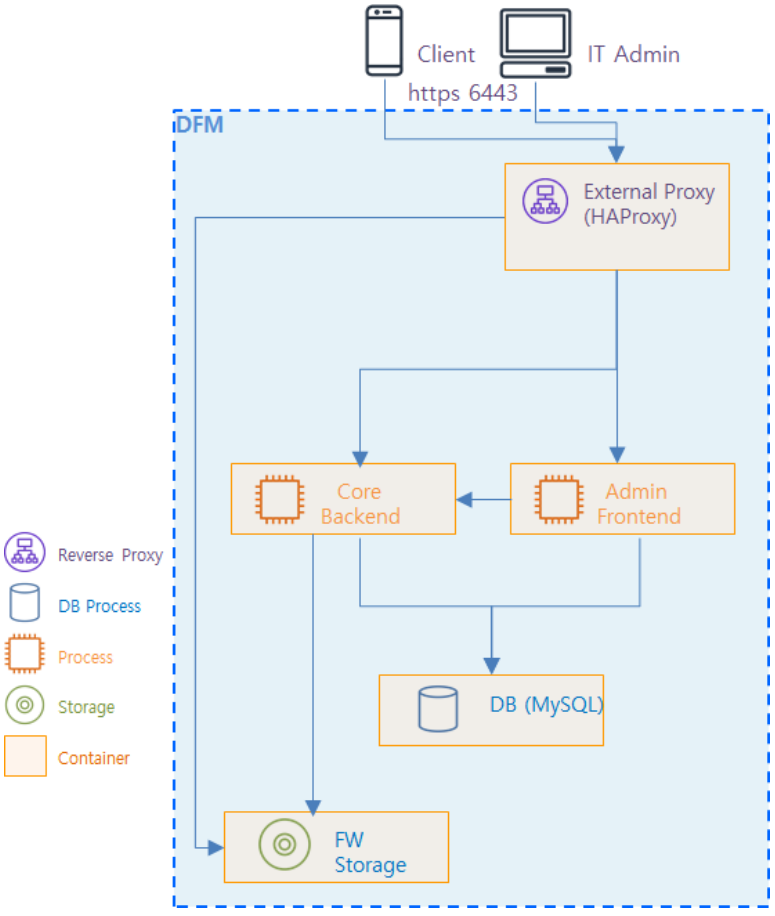


Fig 2-2 Knox E-FOTA On-Premises Product Arch for "Commercial"

3. Deliverables

This chapter describes the actions performed by Samsung to deliver the Knox E-FOTA On-Premises environment.

3.1. DFM Modules

The DFM Module consists of the following core modules:

- **DFM Admin Console Server**: The Frontend module to provide IT admins with an accessible graphical user interface (GUI) on the Google Chrome browser.
- **DFM Core Server**: The Backend module to manage device (client application) actions, integrated into the device using RESTful APIs from the client.
- **DFM Database**: The MySQL-based database contains all service-related data, including device models, their IDs, and policy dependencies in Campaigns.
- **DFM Firmware Storage Management**: The firmware files for downloaded files from the client application.
- **Proxy:** Used for redirection between outer and DFM modules, and for AP Gateway and TLS/SSL termination.

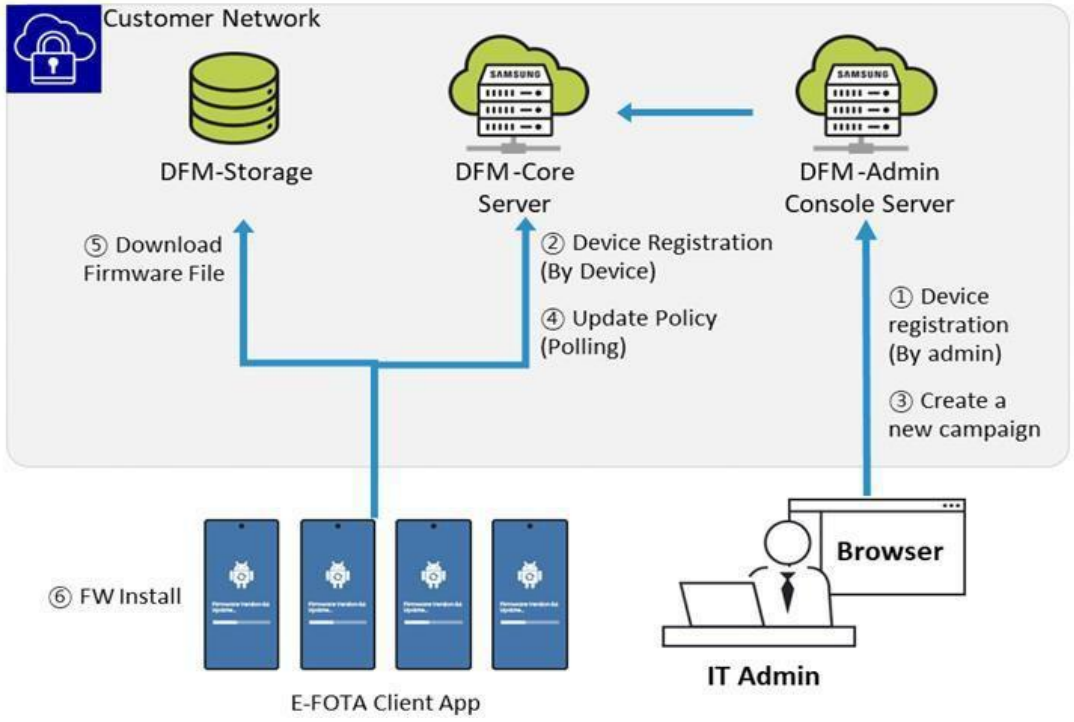


Fig 3-1 **Knox E-FOTA On-Premises Conceptual Architecture**

3.2. Security Considerations

To improve the default security of the Samsung deliverable, it must be implemented using the following standards.

3.2.1. HTTPS and Network encryption

The DFM Module uses HTTPS TLS-based encryption to enhance the security of transactions. The Transport Layer Security (TLS) protocol provides data encryption and verification between applications and servers in scenarios where data is being sent across an insecure network—for example, when working with the DFM Module.

HTTPS header fields are components of the header section of HTTPS request and response messages. They define the operating parameters of a HTTPS transaction. The load balancer and reverse proxy are in front of the DFM Module queries.

3.3. Supported Browser

PLEASE NOTE that this version of the DFM Console UI is designed for **Google Chrome** only.

PART II: Installation and Validation

PART II: Installation and Validation describes how to install the Knox E-FOTA On-Premises service on the customer-provided infrastructure, and how to validate the installed service infrastructure.

4. Installation & Configuration

This chapter explains the first-time installation flow with proper configuration conditions of the DFM Modules. Steps in this chapter run only once during initial installation.
Podman can run in root permission mode and rootless permission mode. If you run in root permission mode, you must attach “sudo” in front of the podman or dfm command. The following document is written based on rootless mode. If a command requires "sudo" because the system is running in root mode, "sudo is required in root mode." will be shown in the document.

4.1. (Prerequisites) Install Package

If the installed Podman version is between 4.0 to 4.4 in Red Hat Enterprise Linux 8.4, 9.2, or 9.6, an additional package installation is required.

The following describes the package that needs to be installed:

podman: manage container tool
podman-plugins: the CNI plugins used to run podman
dnsmasq: use to find local dnsname

【Default】
sudo yum install -y podman
【Red hat 8.4, 9.2, or 9.6 podman 4.0 ~ 4.4】
sudo yum install -y podman podman-plugins dnsmasq

4.2. (STEP01) Create Service Account and Login

The DFM Module is logged in with **a dedicated service account** and operates with the privileges of the account. Therefore, the dedicated service account has to be created in the server. The service account also needs the “**sudo**” privilege as a command permission. Ensure you add your service account into the Wheel group.

We recommend that you create a service account before you start the installation.

The below shows you how to add your service account into the Wheel group:

We assume that you are using the “**nightwatch**” account, and that the DFM Module is logged in with **a dedicated service account** and operates with the privileges of the account.

Add wheel group

sudo usermod --append -G wheel <username>

Example)
sudo usermod --append -G wheel nightwatch

To connect using a created user:

ssh {your-user}@localhost [-p {port}]

Example)
ssh nightwatch@localhost

or if you use port 9000
ssh nightwatch@localhost -p 9000

4.3. (STEP02) Prepare “Disk partition & mount” for DFM modules

The DFM module is installed and operates in the below directory on the **dedicated disk**. Check if the dedicated disk exists and the “partition & mount” is ready, in case the customer has not worked with the disk partition for the DMF module before.

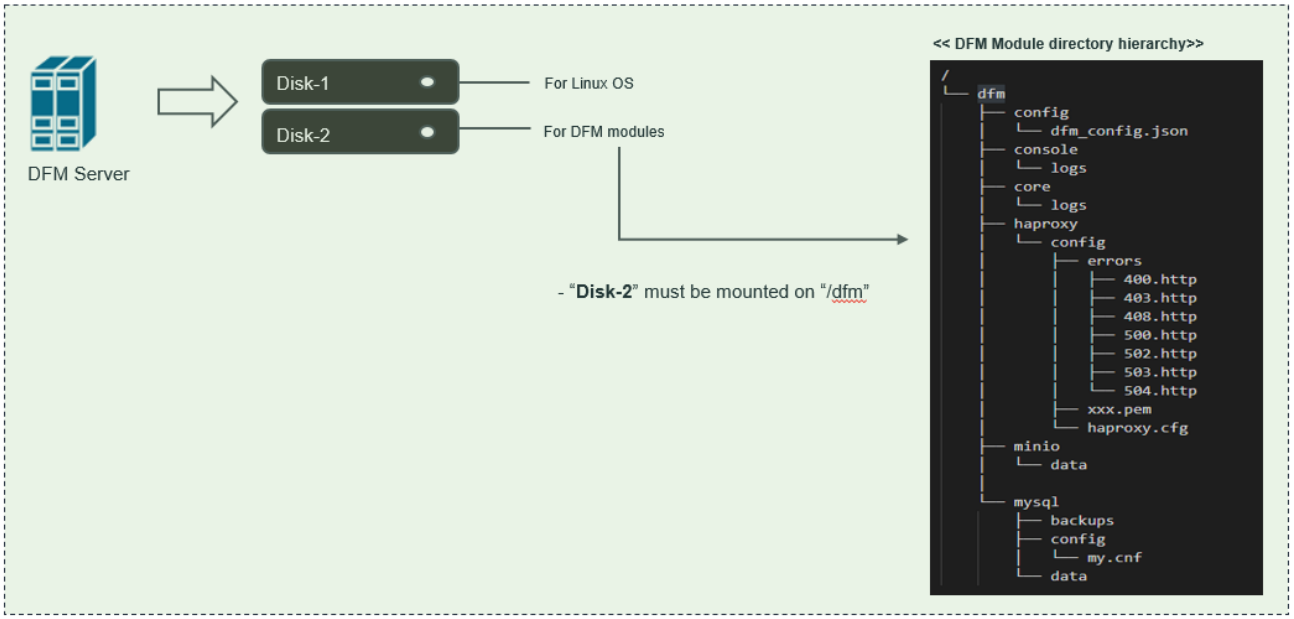


Fig 4-1 An Disk Partitions for DMF Module

For example, we assume that two disks (“sda” and “sdb”) exist.

【CASE01】Disk is Ready

If the disks exist, we don’t need to format and mount them. Now, let’s check the disk information:

sudo lsblk -p

```
NAME         MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
/dev/sda     202:0    0    1T  0 disk
└─/dev/sda1  202:1    0    1T  0 part /
/dev/sdb     202:80   0    1T  0 disk
```

sudo lsblk -f

```
NAME    FSTYPE   LABEL          UUID                                 MOUNTPOINT
sda
└─sda1  ext4     xxxxxxxx-rootfs 6156ec80-9446-4eb1-95e0-9ae6b7a46187 /
sdb     ext4                     d3269ceb-4418-45d0-ba68-d6b906e0595d /dfm
```

⇨ “sdb” is already formatted and mounted on **/dfm**

sudo file -s /dev/sdb

```
/dev/sdb: Linux rev 1.0 ext4 filesystem data, UUID=d3269ceb-4418-45d0-ba68-d6b906e0595d (extents) (64bit) (large files) (huge files)
```

【CASE02】Disk is NOT Ready: it is not formatted

If the disk is not ready, it needs to be formatted and mounted on **/dfm**.

Now, let’s check the disk information:

sudo lsblk -p

```
NAME         MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
/dev/sda     202:0    0    1T  0 disk
└─/dev/sda1  202:1    0    1T  0 part /
/dev/sdb     202:80   0    1T  0 disk
```

sudo lsblk -f

```
NAME    FSTYPE   LABEL          UUID                                 MOUNTPOINT
sda
└─sda1  ext4     xxxxxxxx-rootfs 6156ec80-9446-4eb1-95e0-9ae6b7a46187 /
sdb
```

⇨ “sdb” is NOT formatted

sudo file -s /dev/sdb

```
/dev/sdb: data
```

⇨ This means that the disk needs to be formatted

1) Format with ext4 file-system

```
sudo mkfs –t ext4 /dev/sdb
```

```
sudo mkfs -t ext4 /dev/sdb
mke2fs 1.44.1 (24-Mar-2018)
Creating filesystem with 2621440 4k blocks and 655360 inodes
Filesystem UUID: d3269ceb-4418-45d0-ba68-d6b906e0595d
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

2) Check if the disk is formatted

```
sudo file –s /dev/sdb
```

```
/dev/sdb: Linux rev 1.0 ext4 filesystem data, UUID=d3269ceb-4418-45d0-ba68-d6b906e0595d (extents) (64bit) (large files) (huge files)
```

3) Mount "/dev/sdb" on /dfm

```
// create directory to mount
sudo mkdir /dfm

// mount
sudo mount /dev/sdb /dfm
```

4) Verify

```
df -h
```

```
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb        9.8G   37M  9.3G   1% /dfm
```

【CASE03】Disk is NOT Ready : it is already formatted but not yet mounted on /dfm

If the disk is formatted but not yet mounted, it needs to be mounted on **/dfm**. Now, let's check the disk information:

```
sudo lsblk -p
```

```
NAME         MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
/dev/sda     202:0    0    1T  0 disk
└─/dev/sda1 202:1    0    1T  0 part /
/dev/sdb     202:80   0    1T  0 disk
```

```
sudo lsblk -f
```

```
NAME    FSTYPE   LABEL          UUID                                   MOUNTPOINT
sda
└─sda1  ext4     xxxxxxxx-rootfs 6156ec80-9446-4eb1-95e0-9ae6b7a46187 /
sdb     ext4                     d3269ceb-4418-45d0-ba68-d6b906e0595d
```

⇨ “sdb” ” is formatted but not yet mounted

1) Mount /dev/sdb on /dfm

```
// create directory to mount
sudo mkdir /dfm

// mount
sudo mount /dev/sdb /dfm
```

2) Verify

```
df -h
```

```
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb        9.8G   37M  9.3G   1% /dfm
```

4.3.1 Permanently mount the disk

We recommend that the **customer's IT manager** sets the boot script so that **the dedicated disk** is auto-mounted when the server is booted.

If the **customer's IT manager** has not set the boot script for disk auto-mounting, you should proceed according to the command below.

*If the settings are incorrect, booting may not be possible. The command below is intended for general purposes, and options may differ depending on the customer’s system and situation. Please refer to the “fstab” manual for details.

1) Check mount /dev/sdb on /dfm

```
sudo lsblk -f
```

```
NAME    FSTYPE   LABEL           UUID                                   MOUNTPOINT
sda
└─sda1  ext4     xxxxxxxx-rootfs 6156ec80-9446-4eb1-95e0-9ae6b7a46187 /
sdb     ext4                     d3269ceb-4418-45d0-ba68-d6b906e0595d /dfm
```

2) Edit /etc/fstab file

Please add the contents corresponding to “sdb” to the new line.

```
vi /etc/fstab

UUID=d3269ceb-4418-45d0-ba68-d6b906e0595d     /dfm    ext4   defaults        0 0
```

4.4. (STEP03) Create Service Directories

A separated service directory configuration is required to install and operate the Samsung DFM Module. The service account must have “**read / write / execute**” permissions to the service directory. The service directory should be mounted in a different device location from the OS installation area.

【Service Directory List】

/dfm/mysql/config

* This is where the config file is referenced when the mysql server starts.

/dfm/mysql/data

* This is where databases are created when the mysql server runs.

/dfm/mysql/backups

* This is where databases are backed-up when the mysql server runs.

/dfm/minio/data/dfm-agent-storage

* This is where the E-FOTA client APK files are uploaded when the minio server runs.

/dfm/minio/data/dfm-fw-storage

* This is where firmware binary files are uploaded when the minio server runs.

/dfm/haproxy/config

* This is where the config file is referenced when the haproxy server starts.

/dfm/console/logs

* This is where log files are generated when the admin console server runs.

/dfm/core/logs

* This is where log files are generated when the core server runs.

/dfm/config

* This is where the config file contains the information needed to run the DFM module.

/dfm/background

* This is where the background app file contains the background app for license check.

Now, let's create each service directory.

```
sudo mkdir -p /dfm/mysql/config
sudo mkdir -p /dfm/mysql/data
sudo mkdir -p /dfm/mysql/backups
sudo mkdir -p /dfm/minio/data/dfm-agent-storage
sudo mkdir -p /dfm/minio/data/dfm-fw-storage
sudo mkdir -p /dfm/haproxy/config
sudo mkdir -p /dfm/console/logs
sudo mkdir -p /dfm/core/logs
sudo mkdir -p /dfm/config
sudo mkdir -p /dfm/background
```

Set the service account's permission for the created service directory.

We assume that you are using the "**nightwatch**" account.

```
sudo chown -R nightwatch:nightwatch /dfm
sudo chown -R nightwatch:nightwatch /dfm/mysql
sudo chown -R nightwatch:nightwatch /dfm/minio
sudo chown -R nightwatch:nightwatch /dfm/haproxy
sudo chown -R nightwatch:nightwatch /dfm/mysql/config
sudo chown -R nightwatch:nightwatch /dfm/mysql/data
sudo chown -R nightwatch:nightwatch /dfm/mysql/backups
sudo chown -R nightwatch:nightwatch /dfm/minio/data
sudo chown -R nightwatch:nightwatch /dfm/minio/data/dfm-agent-storage
sudo chown -R nightwatch:nightwatch /dfm/minio/data/dfm-fw-storage
sudo chown -R nightwatch:nightwatch /dfm/haproxy/config
sudo chown -R nightwatch:nightwatch /dfm/console/logs
sudo chown -R nightwatch:nightwatch /dfm/core/logs
sudo chown -R nightwatch:nightwatch /dfm/config
sudo chown –R nightwatch:nightwatch /dfm/background
```

4.5. (STEP04) Install DFM Module Package

The DFM Module is delivered as a tar compress file. This package contains the following resources:

- executable binary (dfm): managed command to run DFM module
- images: podman image about DFM module
- sql query file: DFM module's DB data to initialize mysql
- mysql config file (my.cnf): config file for mysql
- haproxy config file (haproxy.cfg): config file for haproxy
- dfm config file (dfm_config.json): config file for DFM module
- background app files (licenseApp, efota-license.service): background app for license check

- executable binary (dfm):
  - /tmp/sec-dfm_1.0.1.12/dfm/bin/dfm
  - /tmp/sec-dfm_1.0.1.12/dfm/licenseApp
- service excute script:
  - /tmp/sec-dfm_1.0.1.12/dfm/efota-license.service
- images:
  - /tmp/sec-dfm_1.0.1.12/dfm/images/haproxy-debian-2.2.33.tar
  - /tmp/sec-dfm_1.0.1.12/dfm/images/minio-RELEASE.2022-04-30T22-23-53Z.tar
  - /tmp/sec-dfm_1.0.1.12/dfm/images/mysql-8.0.36.tar
  - /tmp/sec-dfm_1.0.1.12/dfm/images/dfm-core_x.x.x.x.tar
  - /tmp/sec-dfm_1.0.1.12/dfm/images/dfm-console_x.x.x.x.tar
- sql query file
  - /tmp/sec-dfm_1.0.1.12/dfm/mysql-query/init_db.sql
  - /tmp/sec-dfm_1.0.1.12/dfm/mysql-query/init_dfm_core.sql
  - /tmp/sec-dfm_1.0.1.12/dfm/mysql-query/init_dfm_console.sql
- mysql config file
  - /tmp/sec-dfm_1.0.1.12/dfm/mysql-config/my.cnf
- haproxy config
  - /tmp/sec-dfm_1.0.1.12/dfm/haproxy-config/haproxy.cfg
- dfm config file
  - /tmp/sec-dfm_1.0.1.12/dfm/dfm_config.json

To extract these resources, the host must unpack the files within the following locations:

The following is a command showing how to extract the package:
If there is already a dfm folder in the /tmp folder, delete it before proceeding.

```
tar -zxvf sec-dfm_{version}.tar.gz -C /tmp

example)
tar -zxvf sec-dfm_1.0.1.12.tar.gz -C /tmp
sec-dfm_1.0.1.12/dfm/
sec-dfm_1.0.1.12/dfm/mysql-query/
sec-dfm_1.0.1.12/dfm/mysql-query/init_dfm_console.sql
sec-dfm_1.0.1.12/dfm/mysql-query/init_dfm_core.sql
sec-dfm_1.0.1.12/dfm/mysql-query/init_db.sql
sec-dfm_1.0.1.12/dfm/dfm_config.json
sec-dfm_1.0.1.12/dfm/mysql-config/
sec-dfm_1.0.1.12/dfm/mysql-config/my.cnf
sec-dfm_1.0.1.12/dfm/haproxy-config/
sec-dfm_1.0.1.12/dfm/haproxy-config/errors/
sec-dfm_1.0.1.12/dfm/haproxy-config/errors/502.http
sec-dfm_1.0.1.12/dfm/haproxy-config/errors/500.http
sec-dfm_1.0.1.12/dfm/haproxy-config/errors/403.http
sec-dfm_1.0.1.12/dfm/haproxy-config/errors/408.http
sec-dfm_1.0.1.12/dfm/haproxy-config/errors/400.http
sec-dfm_1.0.1.12/dfm/haproxy-config/errors/504.http
sec-dfm_1.0.1.12/dfm/haproxy-config/errors/503.http
sec-dfm_1.0.1.12/dfm/haproxy-config/haproxy.cfg
sec-dfm_1.0.1.12/dfm/bin/
sec-dfm_1.0.1.12/dfm/bin/dfm
sec-dfm_1.0.1.12/dfm/images/
sec-dfm_1.0.1.12/dfm/images/minio_mc.tar
sec-dfm_1.0.1.12/dfm/images/mysql-8.0.36.tar
sec-dfm_1.0.1.12/dfm/images haproxy-debian-2.2.33.tar
sec-dfm_1.0.1.12/dfm/images/dfm-core_1.0.1.12.tar
sec-dfm_1.0.1.12/dfm/images/dfm-console_1.0.1.12.tar
sec-dfm_1.0.1.12/dfm/images/minio-RELEASE.2022-04-30T22-23-53Z.tar
```

Next, check if the necessary files exist:

1) check dfm file
ls -l /tmp/sec-dfm_1.0.1.12/dfm/bin/dfm

2) check images
ls -l /tmp/sec-dfm_1.0.1.12/dfm/images
total 969044
-rw-rw-r-- 1 dfm-console_1.0.1.12.tar
-rw-rw-r-- 1 dfm-core_1.0.1.12.tar
-rw-rw-r-- 1 haproxy-debian-2.2.33.tar
-rw-rw-r-- 1 minio-RELEASE.2022-04-30T22-23-53Z.tar
-rw-rw-r-- 1 mysql-8.0.36.tar

3) check sql query file
ls -l /tmp/sec-dfm_1.0.1.12/dfm/mysql-query

total 2076
-rw-r--r-- 1 init_db.sql
-rw-r--r-- 1 init_dfm_console.sql
-rw-r--r-- 1 init_dfm_core.sql

4) check mysql config file
ls -l /tmp/sec-dfm_1.0.1.12/dfm/mysql-config

-rw-rw-r--. 1 my.cnf

5) haproxy config file : haproxy.cfg
ls /tmp/sec-dfm_1.0.1.12/dfm/haproxy-config

haproxy.cfg

6) dfm config file : dfm_config.json
ls /tmp/sec-dfm_1.0.1.12/dfm/dfm_config.json

7) background app files: licenseApp, efota-license.service
ls /tmp/sec-dfm_1.0.1.12/dfm/licenseApp
/tmp/sec-dfm_1.0.1.12/dfm/licenseApp
ls /tmp/sec-dfm_1.0.1.12/dfm/efota-license.service
/tmp/sec-dfm_1.0.1.12/dfm/efota-license.service

4.6. (STEP05) Load Podman Image

Next, register the Podman Images that were unpacked at "**/tmp/sec-dfm_1.0.1.12/dfm/images**". The loaded Podman Images are used when the container is driven. The following shows how to load each Podman Imageusing Podman commands (sudo is required in root mode.):

```
podman load -i /tmp/sec-dfm_1.0.1.12/dfm/images/dfm-console_{version}.tar
podman load -i /tmp/sec-dfm_1.0.1.12/dfm/images/dfm-core_{version}.tar
podman load -i /tmp/sec-dfm_1.0.1.12/dfm/images/haproxy-debian-_{version}.tar
podman load -i /tmp/sec-dfm_1.0.1.12/dfm/images/minio-RELEASE. _{version}.tar
podman load -i /tmp/sec-dfm_1.0.1.12/dfm/images/mysql-_{version}.tar
```

Next, check if the 5 Podman images were loaded. Use the "podman images" command (sudo is required in root mode.):

Example)

```
podman images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
localhost/dfm-console	1.0.1.12	91343f0b589f	2 weeks ago	191 MB
localhost/dfm-core	1.0.1.12	0c838d5b5f1d	2 weeks ago	149 MB
localhost/minio/minio	minio-RELEASE.2022-04-30T22-23-53Z	2f89782ec9dc	3 years ago	57.3 MB
localhost/haproxytech/haproxy-debian	2.2.33	7ba2bc46a616	3 years ago	123 MB
localhost/mysql	8.0.36	f350b0949588	3 years ago	472 MB

4.7. (STEP06) Copy Configuration files

After loading the Podman images, copy the following configuration files into the service directory from the unpacked resources directory.

We assume that you are using the "**nightwatch**" account.

- copy executable binary:

```
// copy executable binary
  sudo cp /tmp/sec-dfm_1.0.1.12/dfm/bin/dfm /usr/bin/ or cp /tmp/sec-
  dfm_1.0.1.12/dfm/bin/dfm /usr/local/bin

// Set executable
sudo chmod 755 /usr/bin/dfm or sudo chmod 755 /usr/local/bin/dfm
```

- **copy mysql config file**:

```
// copy configuration file
  cp /tmp/sec-dfm_1.0.1.12/dfm/mysql-config/my.cnf /dfm/mysql/config

// Set the service account's permission to the configuration file.
  sudo chown -R nightwatch:nightwatch /dfm/mysql/config
```

- **copy haproxy config file**:

```
// copy configuration file
  cp /tmp/sec-dfm_1.0.1.12/dfm/haproxy-config/haproxy.cfg /dfm/haproxy/config

// copy error files
  cp -rf /tmp/sec-dfm_1.0.1.12/dfm/haproxy-config/errors/ /dfm/haproxy/config

//Set the service account's permission to the configuration file.
  sudo chown -R nightwatch:nightwatch /dfm/haproxy/config
```

- **copy dfm config file**:

```
// copy configuration file
  cp /tmp/sec-dfm_1.0.1.12/dfm/dfm_config.json /dfm/config

//Set the service account's permission to the configuration file.
  sudo chown -R nightwatch:nightwatch /dfm/config
```

4.8. (STEP07) Set-up Configuration

In this step, we will set up the initial configuration information needed for the DFM module to run as a Container.

【Configuration List】

- host_ip: Static IP for DFM server.
- listen_port: External listen port at server for DFM module to be accessed.
- listen_scheme: url scheme(http or https) for DFM module to be accessed.
- access_address: domain-based or ip-based
- access_scheme: http or https
- access_port: public port
- public_endpoint: {access_scheme}://{access_address}:{access_port}
- license_app_ip : license app access ip(if you do not need to change ip, the default value is set when input blank, sudo is required in root mode)

In order to properly configure this service after installation, check the customer's network environment in advance. Be sure to check and verify any port-forwarding mapping (NAT) in the customer's network.

Here are a few sample use-cases:

【Use Case 1】IP-based Access Environment

This environment reflects a real-world network environment. The host IP address is not the same, as the public IP address and the CP port number between the public network side and the customer internal network side (including DFM Modules) may be different.

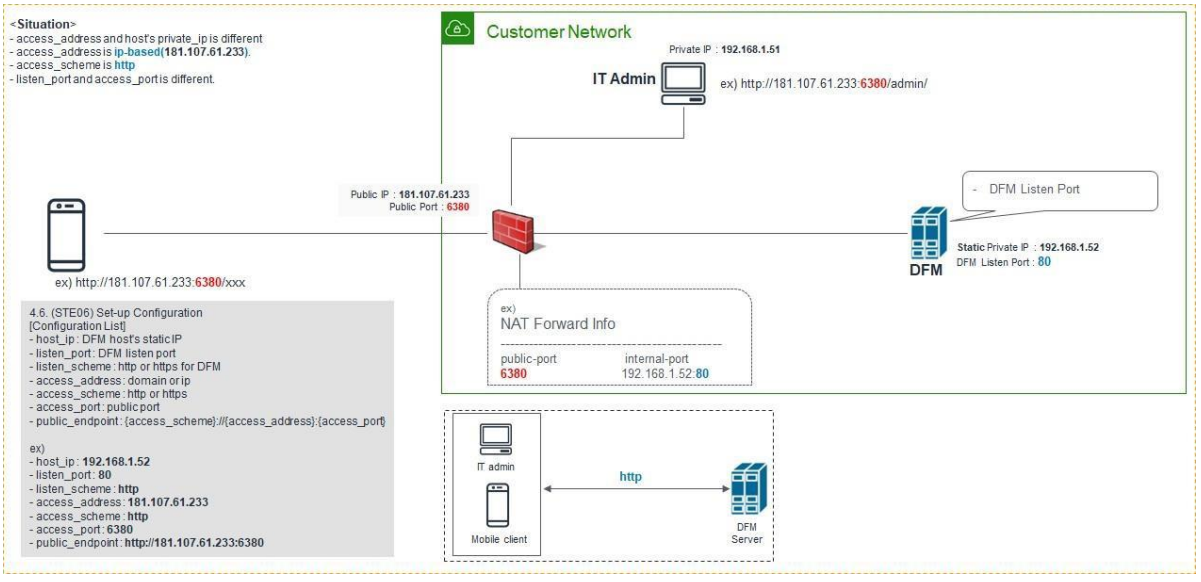


Fig 4-2 IP-based Access Environment

【Use Case 2】Domain-based Access Environment

This environment represents a Domain name-based network environment. You can check the network using the Domain name instead of the IP address.

2-1. (Type A) Using **HTTP**

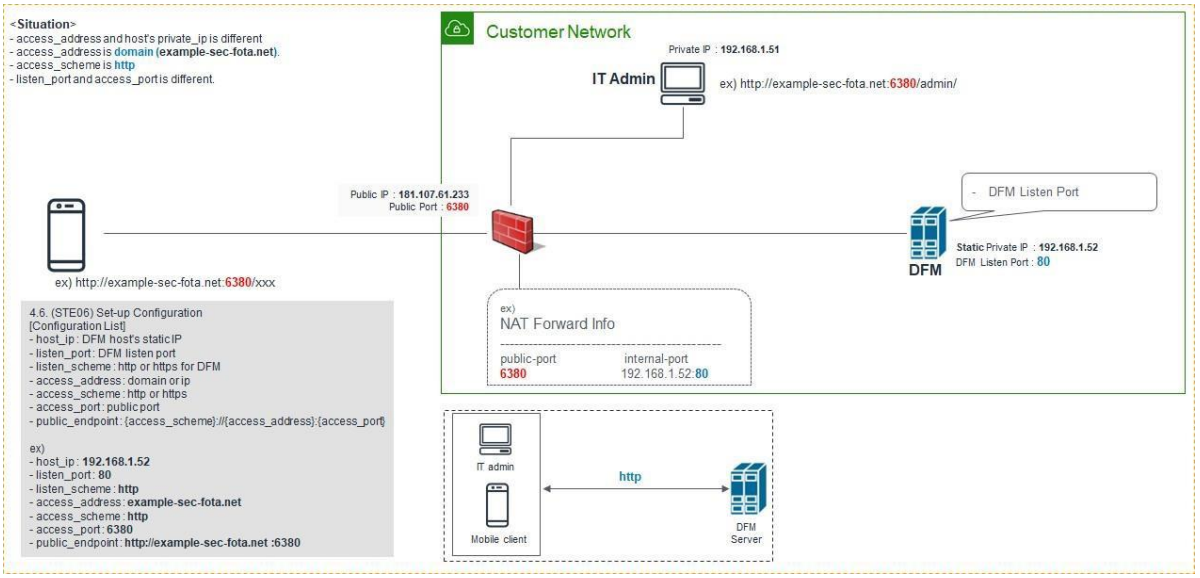


Fig 4-3 **Domain-Based Access Environment (Type A)**

2-2. (Type B) Using **HTTPS** - Customer's LB processes TLS/SSL Termination)

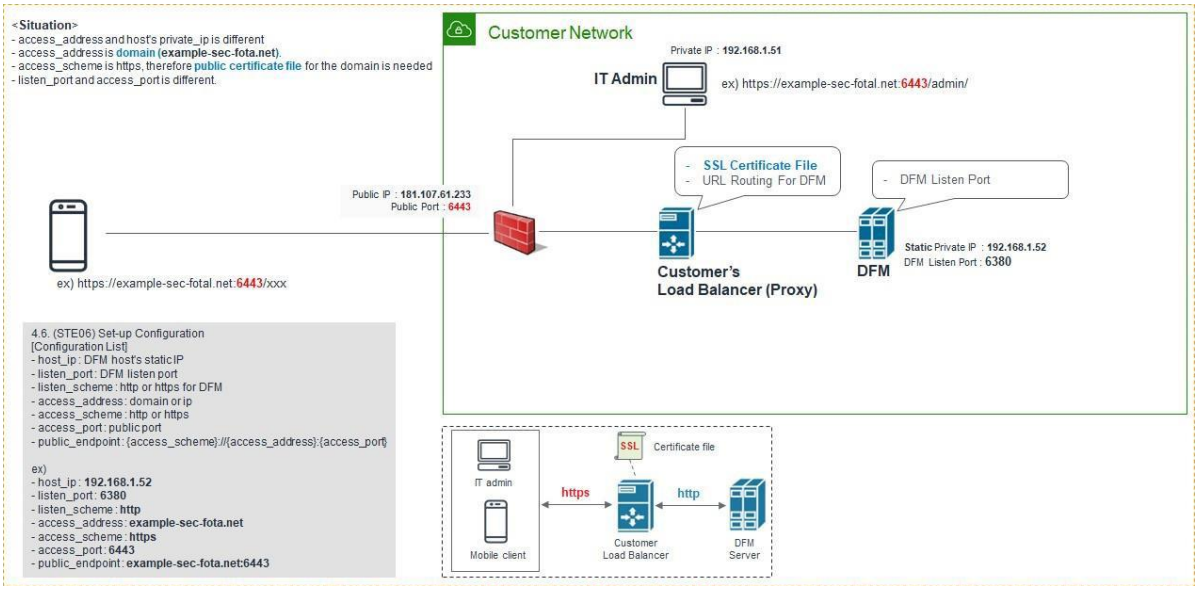


Fig 4-4 **Domain-Based Access Environment (Type B)**

2-3. (Type C) Using **HTTPS** - DFM processes TLS/SSL Termination

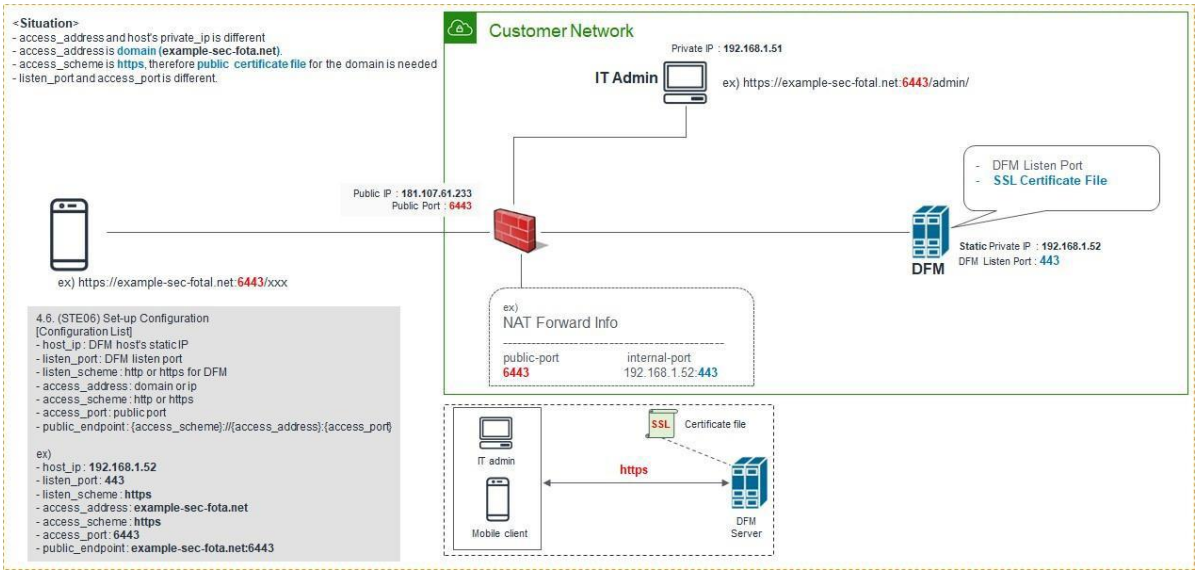


Fig 4-5 **Domain-Based Access Environment (Type C)**

The following is **an example** of how to execute the command to set the above configurations:

2.3.1 (CASE01) IP-Based

- host_ip: 192.168.1.52
- listen_port: 80
- listen_scheme: http
- access_address: 181.107.61.233
- access_scheme: http
- access_port: 6380

The following shows the commands:

```
dfm config set host_ip="192.168.1.52"
dfm config set listen_port="80"
dfm config set listen_scheme="http"
dfm config set access_address="181.107.61.233"
dfm config set access_scheme="http"
dfm config set access_port="6380"
dfm config set license_app_ip=
```

Next, check if the configured value is correct. Use the "**dfm config get {*key*}**" command:

Example)

dfm config get host_ip
192.168.1.52

dfm config get listen_port
80

dfm config get listen_scheme
http

dfm config get access_address
181.107.61.233

dfm config get access_scheme
http

dfm config get access_port
6380

dfm config get license_app_ip
100.0.0.1

2.3.2 (CASE02) Domain-Based

⇦ (**Type ①**) Using **HTTP**

- host_ip: 192.168.1.52
- listen_port: 80
- listen_scheme: http
- access_address: example-sec-fota.net
- access_scheme: http
- access_port: 6380

The following shows the commands:

```
dfm config set host_ip="192.168.1.52"
dfm config set listen_port="80"
dfm config set listen_scheme="http"
dfm config set access_address="example-sec-fota.net"
dfm config set access_scheme="http"
dfm config set access_port="6380"
dfm config set license_app_ip=
```

Next, check if the configured value is correct. Use the "**dfm config get {*key*}**" command:

Example)
dfm config get host_ip
192.168.1.52

dfm config get listen_port
80

dfm config get listen_scheme
http

dfm config get access_address
example-sec-fota.net

dfm config get access_scheme
http

dfm config get access_port
6380

dfm config get license_app_ip
100.0.0.1

⇦ (**Type ②**) Using **HTTPS** - Customer's LB processes TLS/SSL Termination

- host_ip: 192.168.1.52
- listen_port: 6380
- listen_scheme: http
- access_address: example-sec-fota.net
- access_scheme: https
- access_port: 6443

The following shows the commands:

```
dfm config set host_ip="192.168.1.52"
dfm config set listen_port="6380"
dfm config set listen_scheme="http"
dfm config set access_address="example-sec-fota.net"
dfm config set access_scheme="https"
dfm config set access_port="6443"
dfm config set license_app_ip=
```

Next, check if the configured value is correct. Use the "**dfm config get {*key*}**" command:

Example)
dfm config get host_ip
192.168.1.52

dfm config get listen_port
6380

dfm config get listen_scheme
http

dfm config get access_address

example-sec-fota.net

dfm config get access_scheme
https

dfm config get access_port
6443

dfm config get license_app_ip
100.0.0.1

⇦ (**Type ③**) Using **HTTPS** - DFM processes TLS/SSL Termination

- host_ip: 192.168.1.52
- listen_port: 443
- listen_scheme: https
- access_address: example-sec-fota.net
- access_scheme: https
- access_port: 6443

The following shows the commands:

dfm config set host_ip="192.168.1.52"
dfm config set listen_port="443"
dfm config set listen_scheme="https"
dfm config set access_address="example-sec-fota.net"
dfm config set access_scheme="https"
dfm config set access_port="6443"
dfm config set license_app_ip=

Next, check if the configured value is correct. Use the "**dfm config get {*key*}**" command:

Example)
dfm config get host_ip
192.168.1.52

dfm config get listen_port
443

dfm config get listen_scheme
https

dfm config get access_address
example-sec-fota.net

dfm config get access_scheme
https

dfm config get access_port
6443

dfm config get license_app_ip
100.0.0.1

4.8.1. Using the firewalld service

If the firewalld service is in operation on the customer's server, a service port must be added.

```
// List all allowed ports
sudo firewall-cmd --list-ports

// Add a port to the allowed ports to open it for incoming traffic
sudo firewall-cmd --add-port={port number}/tcp

// Make the new settings persistent
sudo firewall-cmd --runtime-to-permanent

Example)
dfm config get listen_port
443
dfm config get access_port
4443
sudo firewall-cmd --add-port=443/tcp --add-port=4443
sudo firewall-cmd --runtime-to-permanent
```

4.8.2. Configure Access port

In Red Hat OS, connection is restricted for ports below 1024. We need to set it to listen_port, which was set above:

```
sudo sysctl net.ipv4.ip_unprivileged_port_start={port number}

Example)
// check listen_port
dfm config get listen_port
443

// open port to listen_port
sudo sysctl net.ipv4.ip_unprivileged_port_start=443
```

4.9. (STEP08) Configure HAProxy

If the external connection type is "**https**", the customer must prepare **1)** the access domain they were issued, **2)** a public certificate for the domain in advance. If the customer is using IP address-based addressing rather than DNS, **this step may be skipped**.

If "ingress_url_scheme" is set to "**https**" on the "***4.7. (STEP07) Set-up Configuration***", this step must be completed.

I. HTTPS Handling

There are two possibilities for TLS/SSL Termination:

1) On Customer's Load Balancer (Proxy)

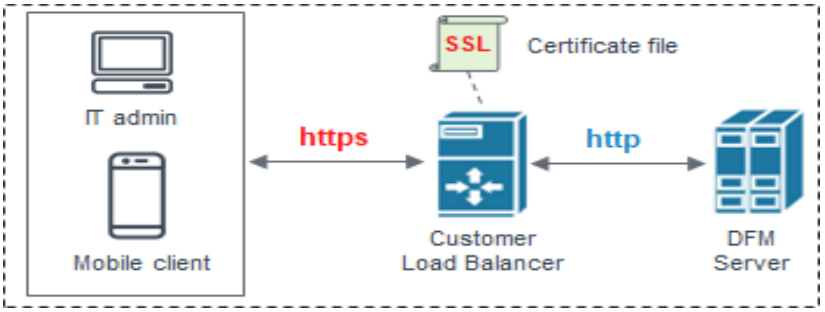


Fig 4-6 **On Customer's Load Balancer (Proxy)**

In this case, the customer's IT manager will operate "public certificate" on its own Load Balancer.

Be careful to comment the "**bind *:443 …**" line and uncomment "**#http-response replace- value Location (.*) https://%[var(txn.host)]/admin/ if logout_path_set**" line in the haproxy.cfg file:

vi /dfm/haproxy/config/haproxy.cfg

```
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
frontend fe_web
    bind *:80
    #bind *:443 ssl crt /usr/local/etc/haproxy/example-sec-fota.net.pem
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
backend dfmConsoleBackend
        mode http
        acl logout_path_set var(txn.path) path /admin/logout
        http-request set-header X-Forwarded-Port %[dst_port]
        http-request add-header X-Forwarded-Proto https if { ssl_fc }

        option httpchk GET /admin/health/live
        http-check expect status 200
        default-server inter 5s fall 3 rise 2

        # if DFM Server is behind customer's Load-Balancer and also customer's Load-Balancer provides ssl termination.
        http-response replace-value Location (.*) https://%[var(txn.host)]/admin/ if logout_path_set
        # otherwise
        #http-response replace-value Location (.*) %[var(txn.scheme)]://%[var(txn.host)]/admin/ if logout_path_set

        server dfm-console dfm-console:10050 check resolvers docker init-addr libc,none
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
```

Since the DFM server can no longer add “Location Header” in response, the **Customer’s Load Balancer must provide the corresponding function**. If the Load Balancer does not provide this function, the user cannot log out after logging into the “admin console webpage” on the DFM.

2) On DFM Server

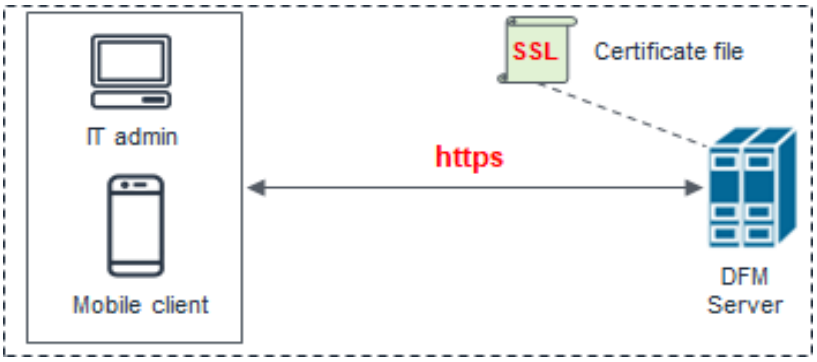


Fig 4-7 On DFM Server

In this case, we need to configure TLS/SSL on our DFM Server. Follow the below steps to do so.

The following assumes that the “**example-sec-fota.net.pem**” file is the public certificate issued by the customer. The public certificate must be copied into haproxy's config folder, and the “haproxy.cfg” file must be edited to change the bind port information and certificate configuration.

- The **crt** parameter identifies the location of the **PEM-formatted** SSL certificate
- **This certificate file** should contain **both the public certificate and private key**
- How to generate the unified certificate for the issued certificate file:

  For example: we assume that you have the below 4 files and the domain's name is **example-sec-fota.net**

  · cert.pem

  · chain.pem

  · fullchain.pem: cert.pem and chain.pem combined

  ⇨ cat cert.pem chain.pem > fullchain.pem

  · privkey.pem

  ⇨ sudo -E bash -c 'cat fullchain.pem privkey.pem > example-sec-fota.net.pem'

  ⇨ ‘**example-sec-fota.net.pem**’ is the unified certificate file

- Also you can make a pem file for devices. Copy the “chain.pem” file to create a new file named “efota.pem”

  ⇨ cp chain.pem efota.pem

We assume that you are using the “**nightwatch**” account.

The certificate file is bound to “/usr/local/etc/haproxy/” in the haproxy container.
So only the certificate file (.pem) name needs to be changed, and not the path.

Be careful to uncomment the "**bind *:443 …**" line and uncomment the "**#http-response replace-value Location (.*) https://%[var(txn.host)]/admin/ if logout_path_set**" line in the haproxy.cfg file:

```
cp example-sec-fota.net.pem /dfm/haproxy/config
sudo chown nightwatch:nightwatch /dfm/haproxy/config/example-sec-fota.net.pem
sudo chmod 600 /dfm/haproxy/config/example-sec-fota.net.pem
vi /dfm/haproxy/config/haproxy.cfg
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

frontend fe_web
        bind *:80
        bind *:443 ssl crt /usr/local/etc/haproxy/example-sec-fota.net.pem
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

backend dfmConsoleBackend
        mode http
        acl logout_path_set var(txn.path) path
        /admin/logout http-request set-header X-
        Forwarded-Port %[dst_port]
        http-request add-header X-Forwarded-Proto https if { ssl_fc }

        option httpchk GET
        /admin/health/livehttp-
        check expect status 200
        default-server inter 5s fall 3 rise 2

        # if DFM Server is behind customer's Load-Balancer and also customer's Load-Balancer provides ssl
        termination.#http-response replace-value Location (.*) https://%[var(txn.host)]/admin/ if
        logout_path_set
        # otherwise
        http-response replace-value Location (.*) %[var(txn.scheme)]://%[var(txn.host)]/admin/ if logout_path_set

        server dfm-console dfm-console:10050 check resolvers docker init-addr libc,none
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
```

3) Mutual TLS

To use mutual TLS, you must create the client certificate and its private key:

1. Create your private key:

```
openssl genrsa -out key.pem 4096
```

2. Create your certificate:

```
openssl req -x509 -new -key key.pem -out cert.pem -days 365
```

3. Encrypt your private key:

```
openssl pkcs8 -topk8 -in key.pem -out key_enc.pem -v2 aes-128-cbc
mv key_enc.pem key.pem
```

4. Combine the two files to create efota_client.pem:

```
cat key.pem cert.pem > efota_client.pem
```

5. Create a copy of cert.pem named client.pem for HAProxy:

```
cp cert.pem client.pem
```

6. Move client.pem to set up HAProxy

```
cp client.pem /dfm/haproxy/config
```

7. Add the settings below to the '2) On DFM Server' settings.

```
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

frontend fe_web
    bind *:80
    bind *:443 ssl crt /usr/local/etc/haproxy/example-sec-fota.net.pem ca-file /usr/local/etc/haproxy/client.pem verify optional

    # monitoring uri
      monitor-uri /health

      http-request capture req.hdr(Host) len 100

      acl acl_dfm_device path_reg ^/dfm/device/v1/*
      acl has_client_cert ssl_fc_has_crt eq 1
      http-request deny if acl_dfm_device !has_client_cert
      use_backend dfmCoreBackend if acl_dfm_device
```

8. To apply the client certificate to your devices, enter the password used to encrypt the private key in the second line of the efota_config file, then push the efota.pem, efota_client.pem, and efota_config files to the Download folder on each of your devices. You can also use the managed configuration. For details, see the Admin guide.

II. HTTP Handling

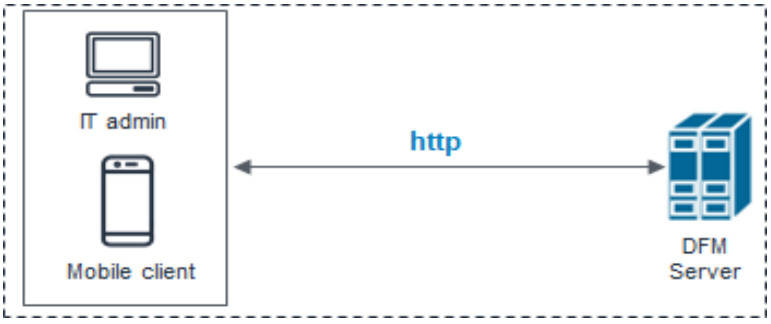


Fig 4-8 On DFM Server

Be careful to comment out the "**bind *:443 …**" line in the haproxy.cfg file in a HTTP-only (non HTTPS) configuration.

【Example】

vi /dfm/haproxy/config/haproxy.cfg

```
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
frontend fe_web
    bind *:80
    #bind *:443 ssl crt /usr/local/etc/haproxy/example-sec-fota.net.pem
```

```
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
backend dfmConsoleBackend
        mode http
        acl logout_path_set var(txn.path) path /admin/logout
        http-request set-header X-Forwarded-Port %[dst_port]
        http-request add-header X-Forwarded-Proto https if { ssl_fc }

        option httpchk GET /admin/health/live
        http-check expect status 200
        default-server inter 5s fall 3 rise 2

        # if DFM Server is behind customer's Load-Balancer and also customer's Load-Balancer provides ssl termination.
        #http-response replace-value Location (.*) https://%[var(txn.host)]/admin/ if logout_path_set
        # otherwise
        http-response replace-value Location (.*) %[var(txn.scheme)]://%[var(txn.host)]/admin/ if logout_path_set

        server dfm-console dfm-console:10050 check resolvers docker init-addr libc,none
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
```

4.10. (STEP09) Create Container Network

The DFM Module is a process executed on a container basis, creating the Podman network required for communications among containers.

To create a network, use the following command (sudo is required in root mode.):

```
dfm network create
```

【Validation】
Run the following command to see if "dfm-network" is visible.

```
dfm network ls

NETWORK ID          NAME          DRIVER
112d0daf3851        dfm-network   bridge
2f259bab93aa        podman        bridge
```

4.11. (STEP10) Copy Background App files

Copy the following Background app files into the service directoryfrom the unpacked resources directory.
We assume that you are using the "**nightwatch**" account.

```
// copy background files
cp /tmp/sec-dfm_1.0.1.12/dfm/licenseApp /dfm/background/licenseApp
cp /tmp/sec-dfm_1.0.1.12/dfm/efota-license.service /etc/systemd/system/efota-license.service
// Set executable
sudo chmod 744 /dfm/background/licenseApp
sudo chcon –t bin_t /dfm/background/licenseApp
```

4.12 (STEP11) Start-up Background App

In this stage, the installer starts the Background App for license check. The command to run the background app is as follows:

```
sudo systemctl daemon-reload
sudo systemctl enable efota-license.service
sudo systemctl start efota-license.service
```

【Validation】
Make sure the Background app is running.

```
sudo systemctl status efota-license.service
Loaded: loaded (/etc/system/system/efota-license.service; enabled; vendor preset: enabled)
Active: active (running) since Tue 2024-XX-XX 06:39:10 UTC; 7s ago
Main PID: 2028 (licenseApp)
```

4.13 (STEP12) Start-up and Initializing the DFM Modules

We have created a service account and signed in using this account to create a service directory, install the DFM Module package, and load the Podman Image resources. The installer has now configured the resources and created the container to network. Now, start up the DFM Modules.

4.13.1 Start-up and Initialize MySQL Server (DFM DB)

In this stage, you will perform the following two steps:

1) Set the DB root password
2) Initialize the SQL query file copied in "4.3 Installing the DFM Module Package" above, on the mysql server

To do this, you must first start the mysql server container.

The command to run the mysql server container is as follows (sudo is required in root mode.):

```
dfm start dfm-mysql
```

【Validation】

Make sure the MySQL container is in a healthy state. It may take some time until its state is healthy.

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-mysql

podman ps -a

CONTAINER ID  IMAGE                      STATUS                         NAMES
2cd1bae13406  localhost/mysql:8.0.36     Up 3 seconds ago (starting)    dfm-mysql

$ podman ps -a

CONTAINER ID  IMAGE                      STATUS                         NAMES
2cd1bae13406  localhost/mysql:8.0.36     Up 52 seconds ago (healthy)    dfm-mysql
```

If the status is healthy, run each of the following commands. (sudo is required in root mode.)

1) Set DB root password: we assume that “password” is “**1q2w3e4r**”
We use this command: ALTER USER 'root'@'localhost' IDENTIFIED BY '{password}'

```
podman exec -it dfm-mysql mysql -uroot

Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 5.7.25-log MySQL Community Server (GPL)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql> ALTER USER 'root'@'localhost' IDENTIFIED BY '1q2w3e4r';
Query OK, 0 rows affected (0.00 sec)

mysql> exit
```

2) Run sql query file : we assume that password is "1q2w3e4r" (sudo is required in root mode.)

```
$ podman exec -i dfm-mysql mysql -uroot -p1q2w3e4r < /tmp/sec-dfm_1.0.1.12/dfm/mysql-query/init_db.sql
mysql: [Warning] Using a password on the command line interface can be insecure.

$ podman exec -i dfm-mysql mysql -uroot -p1q2w3e4r < /tmp/sec-dfm_1.0.1.12/dfm/mysql-query/init_dfm_core.sql
mysql: [Warning] Using a password on the command line interface can be insecure.

$ podman exec -i dfm-mysql mysql -uroot -p1q2w3e4r < /tmp/sec-dfm_1.0.1.12/dfm/mysql-query/init_dfm_console.sql
mysql: [Warning] Using a password on the command line interface can be insecure.
```

4.13.2 Start-up Firmware Storage Server

In this stage, the installer starts the storage server that manages the firmware binary.
The command to run the Firmware Storage Server container is as follows (sudo is required in root mode):

```
dfm start dfm-minio
```

【Validation】

Make sure the Minio container is in a healthy state. It may take some time until its state is healthy.

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-minio

podman ps -a

CONTAINER ID  IMAGE                                                 STATUS                        NAMES
c8f8232a8ca1  localhost/minio/minio:RELEASE.2021-04-18T19-26-29Z    Up 3 seconds ago (starting)   dfm-minio

$ podman ps -a

CONTAINER ID  IMAGE                                                 STATUS                        NAMES
c8f8232a8ca1  localhost/minio/minio:RELEASE.2021-04-18T19-26-29Z    Up 52 seconds ago (healthy)   dfm-minio
```

4.13.3 Start-up DFM Core Server

In this stage, the installer starts the DFM Core Server that provides the core API.
The command to run the core server container is as follows: (sudo is required in root mode.)

```
dfm start dfm-core
```

【Validation】

Make sure the core container is in a healthy state. It may take some time until its state is healthy.

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-core

podman ps -a

CONTAINER ID  IMAGE                          STATUS                        NAMES
120be188f49f  localhost/dfm-core:1.0.1.12     Up 3 seconds ago (starting)   dfm-core

$ podman ps -a

CONTAINER ID  IMAGE                          STATUS                        NAMES
120be188f49f  localhost/dfm-core:1.0.1.12     Up 52 seconds ago (healthy)   dfm-core
```

4.13.4 Start-up DFM Admin Console Server

In this stage, the installer starts the DFM Admin server that provides the admin console web page.
The command to run admin server container is as follows: (sudo is required in root mode.)

```
dfm start dfm-console
```

【Validation】

Make sure the admin container is in a healthy state. It may take some time until its state is healthy.

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-console

podman ps -a

CONTAINER ID  IMAGE                          STATUS                        NAMES
8a3e2f4452e8  localhost/dfm-console:1.0.1.12  Up 3 seconds ago (starting)   dfm-console

$ podman ps -a

CONTAINER ID  IMAGE                          STATUS                        NAMES
8a3e2f4452e8  localhost/dfm-console:1.0.1.12  Up 52 seconds ago (healthy)   dfm-console
```

4.13.5 Start-up HAProxy Server

In this step, the installer starts the HAProxy server that provides TLS/SSL termination and acts as the API gateway.

The command to run HAProxy server container is as follows: (sudo is required in root mode.)

```
dfm start dfm-proxy
```

【Validation】

Make sure HAProxy container is in a healthy state. It may take some time until its state is healthy.

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-proxy

podman ps -a

CONTAINER ID  IMAGE                                          STATUS                        NAMES
e80b80bdba55  localhost/haproxytech/haproxy-debian:2.2.33     Up 3 seconds ago (starting)    dfm-proxy

$ podman ps -a

CONTAINER ID  IMAGE                                          STATUS                        NAMES
e80b80bdba55  localhost/haproxytech/haproxy-debian:2.2.33     Up 52 seconds ago (healthy)    dfm-proxy
```

4.14 How to check Server Operation Status

Finally, the installer has completed the installation of the on-premises service based Podman, and the service is now ready for use. However, we first need to validate whether the above five containers are running in a healthy state.

To check the status of the containers, use the command show below. If every status returns healthy, the service is ready for operation. (sudo is required in root mode.)

```
podman ps -a
```

Example)

```
podman ps -a
CONTAINER ID  IMAGE                                          COMMAND               CREATED      STATUS
c8f8232a8ca1  localhost/minio/minio:RELEASE.2020-06-01T17-28-03Z  server /data          9 hours ago  Up 9 hours ago (healthy)
e80b80bdba55  localhost/haproxytech/haproxy-debian:2.1.4          haproxy -f /usr/l...  8 hours ago  Up 8 hours ago (healthy)
120be188f49f  localhost/dfm-core:1.0.1.4                                                8 hours ago  Up 8 hours ago (healthy)
8a3e2f4452e8  localhost/dfm-console:1.0.1.4                                             8 hours ago  Up 8 hours ago (healthy)
1988a1049bc9  localhost/mysql/enterprise-server:8.0           mysqld                4 hours ago  Up 4 hours ago (healthy)
```

In here, the health status means:

Healthy(0): Normal

Unhealthy(1): Abnormal

Starting (2): Starting

When the installer checks the health status after the installation is completed, if the status is not “Normal”, the installer must redo the installation. If the installation is unsuccessful after several tries, please contact the Samsung engineering team.

We assume that you are using the "**nightwatch**" account.
If you do not enable linger on the account you added dfm — in this case, "nightwatch" — the account will drop the dfm containers and E-FOTA will stop working.
The command to enable linger is:

loginctl enable-linger <username>

Example)
loginctl enable-linger nightwatch

PART III: Initial Operation

PART III describes how to operate the Knox E-FOTA On-Premises service upon completion of the service installation on the customer's infrastructure.

5 Service Operation

This chapter explains how to check the operation status of each DFM Server, and how to use the service properly.

5.1 How to access to admin console page after installing

If you completed every installation step up to "***4.12.5. Start-up HAProxy Server***" in "***4.12. (STEP10) Start- up and Initializing the DFM Modules***", access the admin page to check whether the DFM Service is successfully installed and working.

【URL to the admin site】

{access_scheme}://{access_address}:{access_port}/admin/

⇒ Refer to "***4.9. (STEP07) Set-up Configuration***".

In this guide, we are using the URL and other information as follows:

- host_ip : 192.168.1.52
- listen_port : 80
- listen_scheme : http
- access_address : 181.107.61.233
- access_scheme : http
- access_port : 6380

【Account & Initial Password (PWD)】

⇒ Account will be: **admin**

⇒ Initial PWD will be: **admin12#**

*) *This PWD is created by Samsung, so **change the password** after you sign in.*

【Example】http://192.168.1.52:6380/admin/ (using a new **Chrome** browser)

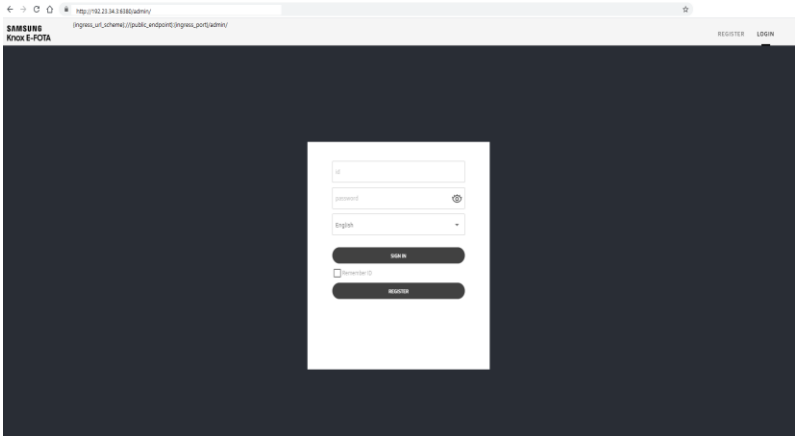


Fig 5-1 The Admin Console for Knox E-FOTA On-Premises

5.2 The Contents Upload

In order to use this service, IT admins must upload the contents (such as license and firmware) properly (please refer to the "**Knox E-FOTA On-Premises User Manual**" provided).

5.3 Troubleshooting and Logging during using the Service

While using this service, any issues should first be addressed on the site to avoid service disruptions from the issues. In order to support issue analysis, Samsung provides the "**TS & Logging Guide for Knox E-FOTA On-Premises**" guide for reference.

5.4 Updating the SSL Certificate when the old certificate is expired

SSL certificates have an expiration date. When the expiration date for the certificate approaches, the customer must reissue the certificate from the certificate signing authority before the current certificate expires.

There are two possibilities for TLS/SSL Termination.

1) On Customer's Load Balancer (proxy)
   We don't need to update the certificate file.
   Refer to (**[Use Case 2]:Type B**) in "***4.9. (STEP07) Set-up Configuration***"

2) On DFM Server
   We need to update the certificate file on the DFM Server.
   Refer to (**[Use Case 2]:Type C**) in "***4.9. (STEP07) Set-up Configuration***"

We assume that the new certificate file is "**new-example-fota.net.pem**", and we also assume that you are using the "**nightwatch**" service account.

【STEP01】Stop Proxy

The command to stop the proxy Server container is as follows: (sudo is required in root mode.)

```
dfm terminate dfm-proxy
```

【STEP02】Copy the new certificate

```
cp new-example-fota.net.pem /dfm/haproxy/config
sudo chown nightwatch:nightwatch /dfm/haproxy/config/new-example-fota.net.pem
sudo chmod 600 /dfm/haproxy/config/new-example-fota.net.pem
vi /dfm/haproxy/config/haproxy.cfg

~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

frontend fe_web
```

```
    bind *:80
    bind *:443 ssl crt /usr/local/etc/haproxy/new-example-sec-fota.net.pem
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
backend dfmConsoleBackend
        mode http
        acl logout_path_set var(txn.path) path /admin/logout
        http-request set-header X-Forwarded-Port %[dst_port]
        http-request add-header X-Forwarded-Proto https if { ssl_fc }

        option httpchk GET /admin/health/live
        http-check expect status 200
        default-server inter 5s fall 3 rise 2

        # if DFM Server is behind customer's Load-Balancer and also customer's Load-Balancer provides ssl termination.
        #http-response replace-value Location (.*) https://%[var(txn.host)]/admin/ if logout_path_set
        # otherwise
        http-response replace-value Location (.*) %[var(txn.scheme)]://%[var(txn.host)]/admin/ if logout_path_set

       server dfm-console dfm-console:10050 check resolvers docker init-addr libc,none
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
```

【STEP03】Restart proxy

The command to restart the proxy Server container is as follows: (sudo is required in root mode.)

```
dfm start dfm-proxy
```

To make sure that the HAProxy container is in a healthy state, run the following command. It may take some time until its state is healthy.

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-proxy

podman ps -a

CONTAINER ID  IMAGE                                       STATUS                          NAMES
e80b80bdba55  localhost/haproxytech/haproxy-debian:2.2.33   Up 3 seconds ago (starting)     dfm-proxy

$ podman ps -a

CONTAINER ID  IMAGE                                       STATUS                          NAMES
e80b80bdba55  localhost/haproxytech/haproxy-debian:2.2.33   Up 52 seconds ago (healthy)     dfm-proxy
```

5.5 DB Backup & Restore.

This section describes how to back up and restore the DB to ensure service continuity.

5.5.1. Back up a MySQL Server Instance

I. ***To BACK UP a MySQL Server instance running in a Podman container using MySQL Shell Backup with Podman:***

1. On the same host where the MySQL Server container is running, start another container with an image of MySQL to perform a backup with the MySQL Shell Backup. Provide access to the server's data directory using the bind mount we created in the last step. Also, mount a host directory (*/path-on-host-machine/backups/* in this example) onto the storage folder for backups in the container (*/backups* in the example) to persist the backups we are creating.

Here is a sample command for this step. Here, we assume that 'root' account's password is **1q2w3e4r** and that the MySql podman image currently installed is localhost/mysql:8.0.36.

We assume that you are using the "**nightwatch**" account. (sudo is required in root mode.)

【Basic Command】

```
podman run --rm --network container:dfm-mysql -v =/path-on-host-machine/datadir/:/backups localhost/mysql:8.0.36
mysqlsh --no-defaults -u{user} -p{password} -e "util.dumpInstance('/backups', {threads: 16, compression: 'zstd'})"
```

【Example】

podman run --rm --network container:dfm-mysql -v /dfm/mysql/backups:/backups localhost/mysql:8.0.36 mysqlsh --no-defaults -uroot -p1q2w3e4r -e "util.dumpInstance('/backups', {threads: 16, compression: 'zstd'})"

```
[nightwatch@xx mysql]$ podman run --rm --network container:dfm-mysql -v /dfm/mysql/backups:/backups
localhost/mysql:8.0.36 mysqlsh --no-defaults -uroot -p1q2w3e4r -e "util.dumpInstance('/backups', {threads: 16,
compression: 'zstd'})"
Cannot set LC_ALL to locale en_US.UTF-8: No such file or directory
WARNING: Using a password on the command line interface can be insecure.
Initializing...
Acquiring global read lock
Global read lock acquired
Initializing - done
Gathering information...
2 out of 6 schemas will be dumped and within them 77 tables, 0 views.
2 out of 5 users will be dumped.
Gathering information - done
All transactions have been started
Locking instance for backup
Global read lock has been released
Writing global DDL files
Writing users DDL
Writing schema metadata...
Writing DDL...
Writing table metadata...
Running data dump using 16 threads.
Dumping data...
NOTE: Table statistics not available for `dfm_console_db`.`tb_system`, chunking operation may be not optimal. Please
consider running 'ANALYZE TABLE `dfm_console_db`.`tb_system`;' first.
NOTE: Table statistics not available for `dfm_console_db`.`databasechangeloglock`, chunking operation may be not
optimal. Please consider running 'ANALYZE TABLE `dfm_console_db`.`databasechangeloglock`;' first.
NOTE: Table statistics not available for `dfm_console_db`.`tb_content_resource`, chunking operation may be not optimal.
Please consider running 'ANALYZE TABLE `dfm_console_db`.`tb_content_resource`;' first.
NOTE: Table statistics not available for `dfm_console_db`.`tb_group`, chunking operation may be not optimal. Please
consider running 'ANALYZE TABLE `dfm_console_db`.`tb_group`;' first.
NOTE: Table statistics not available for `dfm_core_db`.`tczs_lcs_auth`, chunking operation may be not optimal. Please
consider running 'ANALYZE TABLE `dfm_core_db`.`tczs_lcs_auth`;' first.
NOTE: Table statistics not available for `dfm_core_db`.`tczs_dmn`, chunking operation may be not optimal. Please
consider running 'ANALYZE TABLE `dfm_core_db`.`tczs_dmn`;' first.
Writing schema metadata - done
Writing DDL - done
Writing table metadata - done
Starting data dump
Dumping data - done
Dump duration: 00:00:00s
Total duration: 00:00:00s
Schemas dumped: 2
Tables dumped: 77
Uncompressed data size: 499.90 KB
Compressed data size: 58.92 KB
Compression ratio: 8.5
Rows written: 1457
Bytes written: 58.92 KB
Average uncompressed throughput: 499.90 KB/s
   Average compressed throughput: 58.92 KB/s
```

2. The container exits once the backup job is finished and, with the *--rm* option used to start it, it is removed after it exits. An backup files is created and can be found in the host directory mounted in the last step for storing backups:

```
ls /dfm/mysql/backups/
```

5.5.2. Restore a MySQL Server Instance

II. ***To RESTORE a MySQL Server instance in a Podman container using MySQL Shell with Podman:***

1. Stop the MySQL Server container, which also stops the MySQL Server runninginside:
mounted in the last step for storing backups: (sudo is required in root mode.)

```
podman stop dfm-mysql
```

2. On the host, delete all contents in the bind mount for the MySQL Server data directory:

```
sudo rm -rf /dfm/mysql/data/*
```

3. Restart the server container (sudo is required in root mode.)

```
dfm start dfm-mysql
```

4. Set DB root password(we assume that "password" is "**1q2w3e4"**) and and set the 'local_infile' for restore

```
podman exec -it dfm-mysql mysql -uroot -e "ALTER USER 'root'@'localhost' IDENTIFIED BY '1q2w3e4r';"
podman exec -it dfm-mysql mysql -uroot -p1q2w3e4r -e "set global local_infile = 1;"
```

5. Start a container with an image of MySQL to perform the restore with the MySQL Shell. Bind-mount the server's data directory and the storage folder for the backups, like what we did when we backed up the server: (sudo is required in root mode.)

【Basic Command】

```
podman run --rm --network container:dfm-mysql -v /dfm/mysql/backups:/backups localhost/mysql:8.0.36 mysqlsh --no-defaults -u{user} -p{password} -e "util.loadDump('/backups', {threads: 16})"
```

【Example】

```
podman run --rm --network container:dfm-mysql -v /dfm/mysql/backups:/backups localhost/mysql:8.0.36 mysqlsh --no-defaults -uroot -p1q2w3e4r -e "util.loadDump('/backups', {threads: 16})"
[nightwatch@xx ~]$ podman run --rm --network container:dfm-mysql -v
/dfm/mysql/backups:/backups localhost/mysql:8.0.36 mysqlsh --no-defaults -uroot -p1q2w3e4r -e
"util.loadDump('/backups', {threads: 16})"
Cannot set LC_ALL to locale en_US.UTF-8: No such file or directory
WARNING: Using a password on the command line interface can be insecure.
Loading DDL and Data from '/backups' using 16 threads.
Opening dump...
Target is MySQL 8.0.36. Dump was produced from MySQL 8.0.36
Scanning metadata...
Scanning metadata - done
Checking for pre-existing objects...
Executing common preamble SQL
Executing DDL...
Executing DDL - done
Executing view DDL...
Executing view DDL - done
Loading data...
Starting data load
Recreating indexes...
Executing common postamble SQL
Loading data - done
Recreating indexes - done
77 chunks (1.46K rows, 499.90 KB) for 77 tables in 2 schemas were loaded in 0 sec (avg throughput
499.90 KB/s)
0 warnings were reported during the load.
```

The container exits once the backup job is finished and, with the *--rm* option used when starting it, it is removed after it exits.

6. Off the 'local_infile' option

```
podman exec -it dfm-mysql mysql -uroot -p1q2w3e4r -e "set global local_infile = 0;"
```

7. Run init_db.sql for default user registration.

```
podman exec -i dfm-mysql mysql -uroot -p1q2w3e4r < /tmp/dfm/mysql-query/init_db.sql
```

5.6 Configurable length of password digits

The installer can change this default value of a minimum and maximum length of password digits. (default password_min_length=8, default password_max_length=12)

【STEP01】 Stop DFM Admin Console

The command to stop the DFM Admin Console Server container is as follows (sudo is required in root mode.)

```
dfm terminate dfm-console
```

【STEP02】 Set-up the length of the password digits

The minimum length of password is allowed from 8 to 20.
The max length of password is allowed from 12 to 30.

```
dfm config set password_min_length=8
dfm config set password_min_length=20
```

【STEP03】 Check the length of the password digits

```
dfm config get password_min_length
8

dfm config get password_max_length
20
```

【STEP04】 Restart DFM Admin Console

The command to restart the DFM Admin Console Server container is as follows (sudo is required in root mode.)

```
dfm start dfm-console
```

To make sure that the DFM Admin Console container is in a healthy state, run the following command. It maytake some time until its state is healthy.

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-console

podman ps -a

CONTAINER ID  IMAGE                              STATUS                       NAMES
8a3e2f4452e8  localhost/dfm-console:1.0.1.12     Up 3 seconds ago (starting)  dfm-console

$ podman ps -a

CONTAINER ID  IMAGE                              STATUS                       NAMES
8a3e2f4452e8  localhost/dfm-console:1.0.1.12     Up 52 seconds ago (healthy)  dfm-console
```

5.7 Configurable Device Group polling

The installer can change the default value of the device group. (default device_group_enable=false, device_group_max_limit=20000)
This function is used to distribute a large number of devices when serving, and all the devices are distributed across 60 groups.

【STEP01】 Stop DFM Core

The command to stop the DFM Core Server container is as follows: (sudo is required in root mode.)

```
dfm terminate dfm-core
```

【STEP02】 Set up the Device Group polling

The allowed values for “device group enable” are “true” or “false”.
The device group max limit is 20000.

```
dfm config set device_group_enable=true
dfm config set device_group_max_limit=20000
```

【STEP03】 Check the Device Group polling

```
dfm config get device_group_enable
true

dfm config get device_group_max_limit
20000
```

【STEP04】 Restart DFM Core

The command to restart the DFM Core Server container is as follows: (sudo is required in root mode.)

```
dfm start dfm-core
```

To make sure that the DFM Admin Console container is in a healthy state, run the following command. It maytake some time until the state shows as healthy. (sudo is required in root mode.)

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-core

podman ps -a

CONTAINER ID  IMAGE                              STATUS                         NAMES
120be188f49f  localhost/dfm-core:1.0.1.12         Up 3 seconds ago (starting)    dfm-core

$ podman ps -a

CONTAINER ID  IMAGE                              STATUS                         NAMES
120be188f49f  localhost/dfm-core:1.0.1.12         Up 52 seconds ago (healthy)    dfm-core
```

5.8 Configurable device polling interval and postpone waiting time

The installer can change the default value of the device polling interval and postpone the waiting time. (default polling_interval_register=86400, default polling_interval_default=86400, default_waiting_time=30)

【STEP01】 Stop DFM Core

The command to stop the DFM Core Server container is: (sudo is required in root mode.)

```
dfm terminate dfm-core
```

【STEP02】 Set up the device polling interval and postpone waiting time

The polling_interval_register can only be an integer.
The polling_interval_default is allowed from 60 to 86400.(Values lower than 60 will be set to 60)
The postpone waiting time limit is allowed from 1 to 7200.

```
dfm config set polling_interval_register=86400
dfm config set polling_interval_default =86400
dfm config set default_waiting_time=30
```

【STEP03】 Check the device polling interval and postpone waiting time

```
dfm config get polling_interval_register
86400
dfm config get polling_interval_default
86400
dfm config get default_waiting_time
30
```

【STEP04】 Restart DFM Core

The command to restart the DFM Core Server container is: (sudo is required in root mode.)

```
dfm start dfm-core
```

To make sure that the DFM Admin Console container is in a healthy state, run the following command. It may take some time until its state is healthy.

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-core

podman ps -a

CONTAINER ID  IMAGE                          STATUS                         NAMES
120be188f49f  localhost/dfm-core:1.0.1.12     Up 3 seconds ago (starting)    dfm-core

$ podman ps -a

CONTAINER ID  IMAGE                          STATUS                         NAMES
120be188f49f  localhost/dfm-core:1.0.1.12     Up 52 seconds ago (healthy)    dfm-core
```

6. When a Server is Rebooted

This chapter explains the steps to restart the DFM Modules if the server is rebooted, to ensure the service can run properly.

The steps to start the DFM Module server are as follows:

6.1. (STEP01) Log in as the dedicated service account

The DFM Module is logged in with a dedicated service account and operates with the privileges of the account (see, "***4.2. (STEP01) Create Service Account and Login***").

6.2. (STEP02) Prepare "mount" for DFM modules

The DFM module is installed and operates in the below directory on the **dedicated disk**.

The customer **may NOT configure** the auto-mount on the dedicated disk. For such cases, it is necessary to manually mount the dedicated disk on **/dfm**.

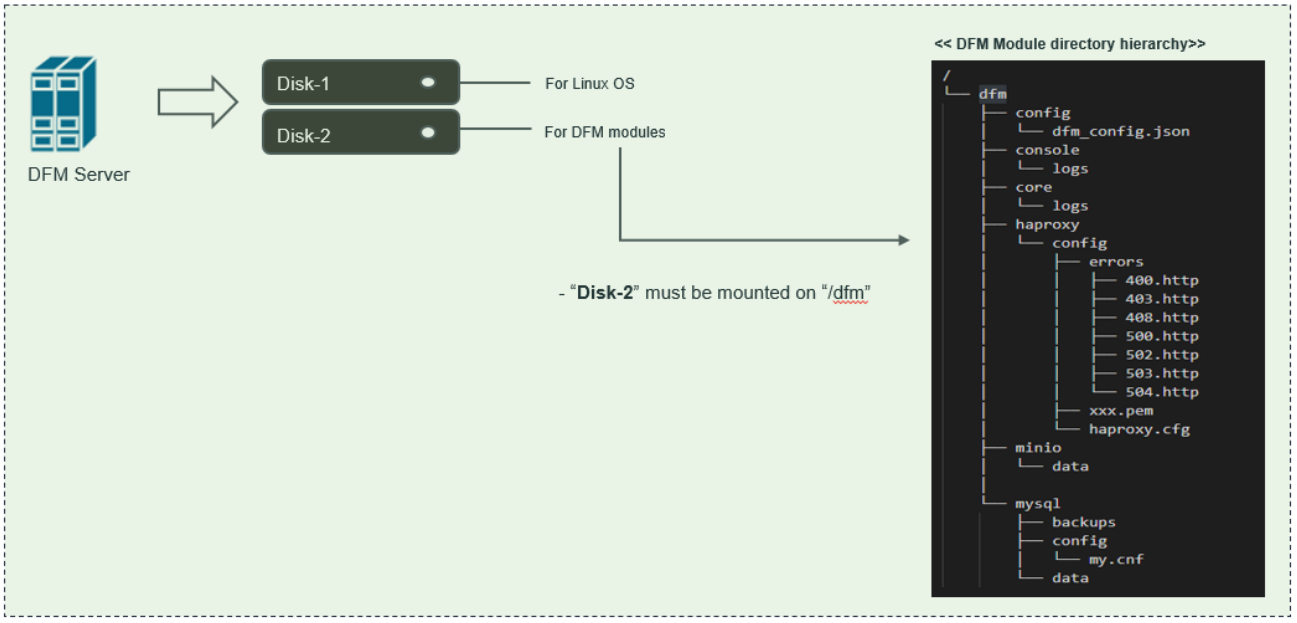


Fig 6-1 **A dedicated disk for DFM module**

For example, we assume that two disks (“sda” and “sdb”) exist.

【CASE01】Disk is Ready

If the disk is ready, we don’t need to mount it. Now, let’s check the disk information:

sudo lsblk –p

```
NAME         MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
/dev/sda     202:0    0    1T  0 disk
└─/dev/sda1 202:1    0    1T  0 part /
/dev/sdb     202:80   0    1T  0 disk
```

sudo lsblk –f

```
NAME    FSTYPE   LABEL          UUID                                   MOUNTPOINT
sda
└─sda1  ext4     xxxxxxxx-rootfs 6156ec80-9446-4eb1-95e0-9ae6b7a46187 /
sdb     ext4                     d3269ceb-4418-45d0-ba68-d6b906e0595d /dfm
```

⇨ “sdb” is already formatted and mounted on **/dfm**

sudo file –s /dev/sdb

```
/dev/sdb: Linux rev 1.0 ext4 filesystem data, UUID=d3269ceb-4418-45d0-ba68-d6b906e0595d (extents) (64bit) (large files) (huge files)
```

【CASE02】Disk is NOT Ready : it is already formatted but Not yet mounted on /dfm

If the disk is formatted but not yet mounted, it needs to be mounted on **/dfm**. Now, let’s check the disk information.

sudo lsblk –p

```
NAME         MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
/dev/sda     202:0    0    1T  0 disk
└─/dev/sda1 202:1    0    1T  0 part /
/dev/sdb     202:80   0    1T  0 disk
```

sudo lsblk –f

```
NAME    FSTYPE   LABEL          UUID                                   MOUNTPOINT
sda
└─sda1  ext4     xxxxxxxx-rootfs 6156ec80-9446-4eb1-95e0-9ae6b7a46187 /
sdb     ext4                     d3269ceb-4418-45d0-ba68-d6b906e0595d
```

⇨ “sdb” ” is formatted but not yet mounted

1) Mount /dev/sdb on /dfm

```
// create directory to mount
sudo mkdir /dfm

// mount
sudo mount /dev/sdb /dfm
```

2) Verify

```
df -h
```

```
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb        9.8G   37M  9.3G   1% /dfm
```

6.3 (STEP03) Start up Database Server (MySQL)

After the system is rebooted, restart MySQL using the following command: (sudo is required in root mode.)

```
dfm restart dfm-mysql
```

【Validation】

Run the following command to ensure the MySOL container is in a healthy state. It may take some time until its state is healthy.

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-mysql

podman ps -a

CONTAINER ID  IMAGE                          STATUS                         NAMES
2cd1bae13406  localhost/mysql:8.0.36         Up 3 seconds ago (starting)    dfm-mysql

$ podman ps -a

CONTAINER ID  IMAGE                          STATUS                         NAMES
2cd1bae13406  localhost/mysql:8.0.36         Up 52 seconds ago (healthy)    dfm-mysql
```

6.4 (STEP04) Start up Firmware Storage Server

After the system is rebooted, restart Minio. The command to run the Minio server container is as follows: (sudo is required in root mode.)

```
dfm restart dfm-minio
```

【Validation】

Run the following command to make sure the Minio container is in a healthy state. It may take some time until its state is healthy.

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-minio

podman ps -a

CONTAINER ID  IMAGE                                          STATUS                       NAMES
c8f8232a8ca1  localhost/minio/minio:RELEASE.2021-04-18T19-26-29Z  Up 3 seconds ago (starting)  dfm-minio

$ podman ps -a

CONTAINER ID  IMAGE                                          STATUS                       NAMES
c8f8232a8ca1  localhost/minio/minio:RELEASE.2021-04-18T19-26-29Z  Up 52 seconds ago (healthy)  dfm-minio
```

6.5 (STEP05) Start up DFM Core Server

After the system is rebooted, restart DFM Core. The command to run the core server container is as follows: (sudo is required in root mode.)

```
dfm restart dfm-core
```

【Validation】

Run the following command to make sure the core container is in a healthy state. It takes some time until its state is healthy.

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-core

podman ps -a

CONTAINER ID  IMAGE                          STATUS                       NAMES
120be188f49f  localhost/dfm-core:1.0.1.12     Up 3 seconds ago (starting)  dfm-core

$ podman ps -a

CONTAINER ID  IMAGE                          STATUS                       NAMES
120be188f49f  localhost/dfm-core:1.0.1.12     Up 52 seconds ago (healthy)  dfm-core
```

6.6 (STEP06) Start up DFM Admin Console Server

After the system is rebooted, restart DFM Admin. The command to run the admin server container is as follows: (sudo is required in root mode.)

```
dfm restart dfm-console
```

【Validation】

Run the following command to ensure the admin container is in a healthy state. It may take some time until its state is healthy.

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-console

podman ps -a

CONTAINER ID  IMAGE                              STATUS                        NAMES
8a3e2f4452e8  localhost/dfm-console:1.0.1.12     Up 3 seconds ago (starting)   dfm-console

$ podman ps -a

CONTAINER ID  IMAGE                              STATUS                        NAMES
8a3e2f4452e8  localhost/dfm-console:1.0.1.12     Up 52 seconds ago (healthy)   dfm-console
```

6.7 (STEP07) Start up HAProxy Server

After the system is rebooted, restart HAProxy. The command to run the HAProxy server container is as follows: (sudo is required in root mode.)

```
dfm restart dfm-proxy
```

【Validation】

Run the following command to ensure HAProxy container is in a healthy state. It may take some time until its state is healthy.

```
# If it is red hat 8.4 version, run health check
podman healthcheck run dfm-proxy

podman ps -a

CONTAINER ID  IMAGE                                          STATUS                        NAMES
e80b80bdba55  localhost/haproxytech/haproxy-debian:2.2.33     Up 3 seconds ago (starting)   dfm-proxy

$ podman ps -a

CONTAINER ID  IMAGE                                          STATUS                        NAMES
e80b80bdba55  localhost/haproxytech/haproxy-debian:2.2.33     Up 52 seconds ago (healthy)   dfm-proxy
```

6.8 (STEP08) Check Server Operation Status

Finally, once every resource is restarted, their states must be verified as healthy. The administrator can use the following command to do so.

If the status of all containers show as healthy, the platform is running normally. (sudo is required in root mode.)

podman ps -a

Example)

podman ps -a

```
CONTAINER ID  IMAGE                                                COMMAND               CREATED       STATUS
c8f8232a8ca1  localhost/minio/minio:RELEASE.2020-06-01T17-28-03Z   server /data          9 hours ago   Up 9 hours ago (healthy)
e80b80bdba55  localhost/haproxytech/haproxy-debian:2.1.4            haproxy -f /usr/l...  8 hours ago   Up 8 hours ago (healthy)
120be188f49f  localhost/dfm-core:1.0.1.4                                                  8 hours ago   Up 8 hours ago (healthy)
8a3e2f4452e8  localhost/dfm-console:1.0.1.4                                               8 hours ago   Up 8 hours ago (healthy)
1988a1049bc9  localhost/mysql/enterprise-server:8.0                 mysqld                4 hours ago   Up 4 hours ago (healthy)
```

PART IV: Update the DFM Modules

PART IV: Update the DFM Modules describes how to update the Knox E-FOTA On-Premises service if there are any updates within the service resources.

7. Update the DFM Module

This chapter explains how to update the DFM Modules in operation, such as a fetch version. In order to properly update each module, the updater must first stop the module based on the related command (see, ***Appendix B***).

During the update, the Knox E-FOTA On-Premises service may not be available.

The DFM Module is logged in with a dedicated service account and operates with the privileges of the account. Ensure you log in with the account you previously used for installation.

7.1. Podman Image Update

If there is an updated DFM Module, it is also released as a Podman Image Package and packed as a tar file. In the release, the Podman Image contains repository and tag information as well.

7.1.1. DFM Database Update (MySQL)

For example, assume that the released **MySQL** image information is as follows:

- Podman image: mysql- xx.xx.xx.tar
- repository: localhost/mysql
- tag: xx.xx.xx

It should be updated as per the following steps: (sudo is required in root mode.)

【**STEP01**】Stop the running DFM Core Server, Admin Console Server, and Mysql Server.

```
dfm terminate dfm-core
dfm terminate dfm-console
dfm terminate dfm-mysql
```

【**STEP02**】Load the released Podman Image.

```
podman load -i mysql- xx.xx.xx.tar
```

【**STEP03**】Change the repository and tag's configuration

```
dfm config set mysql_img_rep=localhost/mysql
dfm config set mysql_img_tag=xx.xx.xx
```

【**STEP04**】Confirm the changed repository and tag's configuration

```
dfm config get mysql_img_rep
dfm config get mysql_img_tag
```

【**STEP05**】Start up Server

- MySQL Server

```
dfm start dfm-mysql
```

【Validation】

Run the following command to ensure the mysql container is in a healthy state. It takes some time until its state is healthy.

```
podman ps -a
```

7.1.2. DFM Firmware Storage Update (MinIO)

For example, assume that the released **MinIO** image information is as follows: (sudo is required in root mode.)

- Podman image : minio-xx.xx.xx.tar
- repository : localhost/minio/minio
- tag : xx.xx.xx

【**STEP01**】Stop the MinIO server.

```
dfm terminate dfm-minio
```

【**STEP02**】Load the released Podman Image.

```
podman load -i minio-xx.xx.xx.tar
```

【**STEP03**】Change the repository and tag's configuration

```
dfm config set minio_img_rep=localhost/minio/minio
dfm config set minio_img_tag=xx.xx.xx
```

【**STEP04**】Confirm the changed repository and tag's configuration

```
dfm config get minio_img_rep
dfm config get minio_img_tag
```

【**STEP05**】Start-up Server

- MinIO Server

```
dfm start dfm-minio
```

【Validation】
Run the following command to ensure the mysql container is in a healthy state. It takes some time until its state is healthy.

```
podman ps -a
```

7.1.3. DFM Core Update

For example, assume that the released **Core** image information is as follows: (sudo is required in root mode.)

- Podman image : dfm-core-xx.xx.xx.tar
- repository : localhost/dfm-core
- tag : xx.xx.xx

【**STEP01**】Stop the running core server.

```
dfm terminate dfm-core
```

【**STEP02**】Load the released Podman image.

```
podman load -i dfm-core-xx.xx.xx.tar
```

【**STEP03**】Change the repository and tag's configuration

```
dfm config set core_img_rep=localhost/dfm-core
dfm config set core_img_tag=xx.xx.xx
```

【**STEP04**】Confirm the changed repository and tag's configuration

```
dfm config get core_img_rep
dfm config get core_img_tag
```

【**STEP05**】Start-up Server

- DFM Core Server

```
dfm start dfm-core
```

【Validation】

Run the following command to ensure the mysql container is in a healthy state. It takes some time until its state is healthy.

```
podman ps -a
```

7.1.4. DFM Admin Console Update

For example, assume that the released **Admin** image information is as follows: (sudo is required in root mode.)

- Podman image : dfm-console-xx.xx.xx.tar
- repository : localhost/dfm-console
- tag : xx.xx.xx

【**STEP01**】Stop the running core, admin and mysql servers.

```
dfm terminate dfm-console
```

【**STEP02**】Load the released Podman image.

```
podman load -i dfm-console-xx.xx.xx.tar
```

【**STEP03**】Change the repository and tag's configuration

```
dfm config set console_img_rep=localhost/dfm-console
dfm config set console_img_tag=xx.xx.xx
```

【**STEP04**】Confirm the changed repository and tag's configuration

```
dfm config get console_img_rep
dfm config get console_img_tag
```

【**STEP05**】Start-up Server

- Admin Console Server

```
dfm start dfm-console
```

【Validation】
Run the following command to make sure the mysql container is in a healthy state. It takes some time until its state is healthy.

```
podman ps -a
```

7.1.5. HAProxy update

For example, assume that the released **HAProxy** image information is as follows: (sudo is required in root mode.)

- Podman image : haproxy-debian-xx.xx.xx.tar
- repository : localhost/haproxytech/haproxy-debian
- tag : xx.xx.xx

【**STEP01**】Stop the running haproxy server.

```
dfm terminate dfm-proxy
```

【**STEP02**】Load the released Podman image.

```
podman load -i haproxy-debian-xx.xx.xx.tar
```

【**STEP03**】Change the repository and tag's configuration

```
dfm config set haproxy_img_rep=localhost/haproxytech/haproxy-debian
dfm config set haproxy_img_tag=xx.xx.xx
```

【**STEP04**】Confirm the changed repository and tag's configuration

```
dfm config get haproxy_img_rep
dfm config get haproxy_img_tag
```

【**STEP05**】Start-up Server

- HAProxy Server

```
dfm start dfm-proxy
```

【Validation】
Run the following command to ensure the HAProxy container is in a healthy state. It may take some time until its state is healthy.

```
podman ps -a
```

7.2. The Contents Update

In order to use this service, IT admins must upload the contents (such as license and firmware) properly (please refer to the "**Knox E-FOTA On-Premises User Manual**" provided).

PART V: Purge DFM Modules

This section, which covers purging the DFM Modules, describes how to erase all installed services when you want to delete the existing installed modules.

Please note that doing so **erases all existing data**.

After completing these actions, you can reinstall the DFM modules without any interference from the old installation (see 4.3. (STEP03) Create Service Directories).

8. Purge the DFM Modules

This chapter explains how to purge the installed DFM Modules.

The DFM Module is logged in with a dedicated service account and operates with the privileges of the account. Log in with the account you used during the installation.

8.1. Terminate Services

If there are active services, terminate them. (sudo is required in root mode.)

【STEP01】 Check if there are any running or exited services. If they exist, we need to terminate them.

podman ps –a

Example)

podman ps –a

```
CONTAINER ID  IMAGE                                          COMMAND               CREATED      STATUS
97e26cc3bea3  registry.access.redhat.com/ubi8/pause:latest                          9 days ago   Up 9 days ago
87788c0f949a  localhost/mysql/enterprise-server:8.0           mysqld                9 days ago   Up 9 days ago
4f6bb6af2920  localhost/dfm-console:1.0.1.2-rootless                                9 days ago   Up 9 days ago
636ab5081c12  localhost/dfm-core:1.0.1.2-rootless                                   9 days ago   Up 9 days ago
0f9bc568fcd5  localhost/minio/minio:RELEASE.2020-06-01T17-28-03Z  server /data          9 days ago   Up 9 days ago
af2c052532d5  localhost/haproxytech/haproxy-debian:2.1.4        haproxy -f /usr/l...  9 days ago   Up 9 days ago
```

1. DFM Database (MySQL)

   Stop the server with the following command:

   dfm terminate dfm-mysql

2. DFM Firmware Storage (MinIO)

   Stop the server with the following command:

   dfm terminate dfm-minio

3. DFM Core Server

   Stop the server with the following command:

   dfm terminate dfm-core

4. DFM Admin Console Server

   Stop the server with the following command:

   dfm terminate dfm-console

5. DFM HAProxy Server

   Stop the server with the following command:

   dfm terminate dfm-proxy

6. Check if all services are removed.

   Check with the following command:

   podman ps -a

7. Stop Background App and check

Stop background app with the following command:

```
sudo systemctl stop efota-license.service
```

Check with the following command:

```
sudo systemctl status efota-license.service
  Loaded: loaded (/etc/system/system/efota-license.service; enabled; vendor preset: enabled)
Active: inactive (dead) since Tue 2024-XX-XX 06:39:10 UTC; 7s ago
```

8.2. Remove Service directory

Remove old data using the following:

Remove all directory in /dfm

```
cd /dfm
sudo rm -rf *
```

PART VI: APPENDICES

PART IV: APPENDICES presents more in-depth explanations for each item.

APPENDICES

Appendix A. Terms and Abbreviations

This chapter outlines the terms and abbreviations used in this guide.

App: Application
CAT: Category Codes
CSO/TEO: Customer Service Operation/Technical Engineer for On-Premise
CM: Commercial Type Product
DE: Docker Enterprise
DFM: Device Firmware Management
DNS: Domain Name Server
E2E: End to End
E-FOTA: Enterprise – Firmware over the Air
FYI: For Your Information
HA: High Availability
H/W: Hardware
ID: Identification
KE: Knox E-FOTA (Brand)
LB: Load Balancer
NAT: Network Address Translation
OS: Operating System
PoC: Proof of Concept
PWD: Password
SSL: Secure Sockets Layer
TLS: Transport Layer Security, successor to SSL
UI: User Interface

Appendix B. How to terminate each DFM Module

These commands should not be used in normal operation, as stopping a module can seriously impact how the service runs. Use this command for updates, such as when there is a fetch version delivery. (sudo is required in root mode.)

1. DFM Database (MySQL)

   Stop the server with the following command:

   ```
   dfm terminate dfm-mysql
   ```

2. DFM Firmware Storage (MinIO)

   Stop the server with the following command:

   ```
   dfm terminate dfm-minio
   ```

3. DFM Core Server

   Stop the server with the following command:

   ```
   dfm terminate dfm-core
   ```

4. DFM Admin Console Server

   Stop the server with the following command:

   ```
   dfm terminate dfm-console
   ```

5. DFM HAProxy Server

   Stop the server with the following command:

   ```
   dfm terminate dfm-proxy
   ```

Appendix C. Summary for Software (S/W) Recommendation

Read more about detailed recommendations in "*2.3. Recommendation Per each Product usage*".

Product	Category	S/W	Version	Supported Options	Additional Info
CM	**Server OS**	Red hat	8.4, 9.2, or 9.6	Enterprise (**Paid**)	
	Container	Podman	over 4.0		
	Database	MySQL	Community Edition	Community (Free)	
PoC	**Server OS**	Red hat	8.4, 9.2, or 9.6	Community (free)	
	Container	Podman	Over 4.0		
	Database	MySQL	Community Edition	Community (Free)	

Appendix D. A Recommended Schedule for On-Site Installation by CSO/TEO

This recommended schedule can be used by the CSO/TEO while they are doing the on-site installation. The detailed schedule can be freely modified.

We recommend "The 4-Day Installation", as the customer should understand how they are using the Knox E-FOTA On-Premises service during this program. A training session should be included to support this purpose as well.

Day	Actions	Program
Day1	Check the customer's infrastructures (such as H/W and S/W) to install the service on, based on the prerequisites (see "*2.3 Recommendation Per each Product usage*")	1. Introduce each other 2. Introduce "The 4-Days Installation" program 3. Introduce the Knox E-FOTA On-Premises service (using "**KE On-Premise Service Intro 2020.pdf**" 4. Check the customer's infrastructures 1) H/W recommendation, such as Server CPU cores, RAM, Disk, Network Card 2) S/W recommendation, such as Operating System, Docker Engine, MySQL Edition, and whether those have been installed by the customer 3) Get public certificate files for https 4) Get port number (6443) for https 5. Wrap-up
Day2	Perform the installation based on this guide (see "*4. Installation & Configuration*")	1. Introduce the program to install 2. Start Installation 3. Configure the DFM service infrastructure 4. Check the service operation via the Web Console 5. Wrap-up
Day3	Perform an acceptance test through E2E with devices	1. Introduce how to do an E2E test with devices 2. Introduce how to use the service Web Console (using "**Knox E-FOTA On-Premises User Guide.pdf**, and **Knox E-FOTA On-Premises User Guide for Device.pdf**") 3. Upload the License onto the Server 4. Upload the Firmware deltas (Contents for FOTA) 5. Upload the device information used during the test 6. Create the Campaign 7. Do E2E test with devices 8. Wrap-up
Day4	Introduce Operation and Maintenance procedures (Get document for "**The Confirmation of Installation Process End**" from the Customer)	1. Introduce the steps and how to perform them if there is an issue ☜ Using "**TS & Logging Guide for Knox E-FOTA On-Premise.pdf**" 2. Introduce how to raise issues ☜ Using "**Issue raising process**" 3. Introduce service operation steps 4. ☜ Using "**Service Operation Guide**" 5. Sign the "Notice for Completion Installation" ☜ Refer to "**Appendix E**" (*Installation and Initial Operation Guide for Knox E-FOTA On-Premises.pdf*) 6. Wrap-up

Appendix E. An Example of "Notice for Completion Installation"

Notice for Completion Installation

Dear < Customer Name >,

This form is to sign-off completion of your project with us. Kindly complete as best as possible and send back to us.

PRODUCT: Knox E-FOTA One On-premise	MANAGER NAME:
START DATE:	COMPLETION DATE:
June 1 2020 ~ June 4 2020	

DELIVERABLES:

1. **Device Client**
   It means Client application running on Samsung mobile devices. It is responsible for interacting with the E-FOTA (Enterprise-Firmware Over The Air) Server, including binary package download, and installer activation for the binary package.

2. **Device Firmware Management (DFM)**
   It is a main module for E-FOTA, including managed devices to FOTA, creation and management of FOTA Campaigns, and Firmware binaries for devices.
   It is consist of followings:
   1) DFM Core – It consists of Core Backend and Front End for Administrators
   2) DB (MySQL) – It is a data base for system operation
   3) Storage – It is a storage for Firmware binaries

3. **Installed in Customer's Environment**
   It depends on the contraction.
   1) Pre-Prod Environment (1 Set)
   2) Prod Environment (1 Set)

CUSTOMER'S COMMENTS:

REMARK:

By signing this document, I acknowledge that I have delivered all the stated deliverables.	By signing this document, I acknowledge that I have received all the stated deliverables.
Samsung (subsidiary office name)	**< Customer Name >**
Name:_____	Name:_____
Signature:_____	Signature:_____
Date: _____	Date: _____

We recommend that you complete and send this form within 5 working days. However, if after this period we do not receive the completed form, we shall assume that the project has been signed off by you and no further action will be required of you.

Appendix F. Set E-FOTA agent config by managed Configuration

The Knox E-FOTA On-Premises agent requires your server URL and TLS certificate to connect to your on-premises server. The Knox E-FOTA On-Premises agent supports a managed configuration which lets you configure connection and authentication details needed for the app to communicate with your on-premises server. Follow your EMM's documentation on how to configure an app's managed configuration.

Reference: https://developer.android.com/work/managed-configurations

The managed configuration has the following fields:

Field	Description
server_url	*string* The URL of your Knox E-FOTA On-Premises server
pem	*string* Your server's TLS certificate
client_cert_pem	*string* (Only used for mutual TLS) Client certificate used by clients to authenticate themselves to your server
client_key_pem	*string* (Only used for mutual TLS) Encrypted private key used by clients for signing
client_pem_password	*string* (Only used for mutual TLS) Password for the encrypted private key

< EOF (End Of File) >